LLM-Based Annotation Quality Enhancement for Human Trafficking Data

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Abstract

High-quality labelled data is a basic requirement for supervised NLP, but in domains like MSHT it is rarely clean. Annotation is expensive, inconsistent, and dependent on expertise that is hard to find. This dissertation examines what happens when you take those problems seriously in a real-world dataset of 2,800 English and Spanish articles on Modern Slavery and Human Trafficking, annotated through a crowdsourced pipeline by Stop the Traffik.

Exploratory data analysis found that 98.2% of tasks had been labelled by a single annotator, that measured inter-annotator agreement averaged just 26.1% across the small subset of tasks with multiple annotators, and that nearly all annotations fell into the LOW confidence tier of a weighted quality scoring formula accounting for annotator count, agreement, and completion time. Agreement-based quality assessment was therefore unviable. Additionally, 969 annotations were flagged as suspiciously fast, some completed in under two seconds for articles of several hundred words. These findings ruled out conventional quality remedies and pointed to a different approach: LLM-based annotation enhancement without reliance on a verified human reference.

Three established frameworks (Active-Prompt, LLMaAA, and Reflexion) were implemented as LangGraph pipelines and evaluated across Named Entity Recognition (NER) and Natural Language Classification (NLC) tasks in both languages. All three handled NLC reliably but failed on NER in distinct ways: LLMaAA produced a 0% confidence upgrade rate; Reflexion verified only 30% of English NER tasks across three self-critique trials; and Active-Prompt identified high uncertainty across all NER tasks but produced no resolved annotation. These failures traced to a shared root cause: the combinatorial label space of the NER schema makes structural inter-sample disagreement the default condition, not an edge case that iterative correction can fix.

Active-Synthesis was designed in response. The framework runs five parallel GPT-4o sampling calls, scores outputs using disagreement count and Shannon entropy to identify conflicting cases and resolves conflicts by presenting all five candidate annotations simultaneously to the model and requesting a single justified final label. Active Synthesis resolved 70.7% of English NER tasks and 80.5% of Spanish NER tasks, maintaining 100% NLC approval. Escalated outputs are not returned as failures: each comes with a candidate annotation and a structured rationale identifying the specific conflicts that required human judgement. Active-Synthesis represents a novel framework that combines uncertainty-based parallel sampling with simultaneous multi-sample synthesis for annotation conflict resolution in a specialised NLP domain.

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List of Abbreviations

Abbreviation	Full Form
NER	Named Entity Recognition
NLC	Natural Language Classification
LLM	Large Language Model
IAA	Inter-Annotator Agreement
EDA	Exploratory Data Analysis
API	Application Programming Interface
ML	Machine Learning
NLP	Natural Language Processing
MSHT	Modern Slavery and Human Trafficking
CoT	Chain-of-Thought
KNN	K-Nearest Neighbours
PII	Personally Identifiable Information
BERT	Bidirectional Encoder Representations from Transformers
SLM	Small Language Model
GPT	Generative Pre-trained Transformer
F1	F1 Score (Harmonic Mean of Precision and Recall)
JSON	JavaScript Object Notation
LangGraph	Graph-based LLM Orchestration Framework
ABSTAIN	Annotation Label Indicating No Applicable Category
BADGE	Batch Active learning by Diverse Gradient Embeddings
GMM	Gaussian Mixture Model
GPU	Graphics Processing Unit
NGO	Non-Governmental Organisation

Chapter 1: Introduction

1.1 Background and Problem Statement

Modern slavery and human trafficking (MSHT) affect an estimated 40 million people worldwide yet remain chronically underreported and difficult to detect at scale. Stop The Traffik (STT), an NGO operating across more than 50 countries, addresses this through its STOP app: a crowdsourced intelligence platform that collects public reports of suspected trafficking activity. Each report is assigned one or more named entity and classification labels by human annotators; the result is a structured dataset intended for downstream analysis and model training.

The dataset has a structural problem. A preliminary exploratory data analysis found that 98.2% of all annotations carry a single annotator, with a mean inter-annotator agreement of just 26.1%. A further 969 annotations were flagged as speed-outliers (completed in under 30 seconds), which points to low-effort labelling. These conditions produce ground truth that is sparse, inconsistent, and likely incomplete: the kind of label noise that propagates directly into any model trained on it.

LLM-based annotation has emerged as a viable route to scaling annotation quality, but the existing frameworks were not designed for this context. Datasets with single annotators provide no reliable external standard against which LLM outputs can be calibrated. A 58-type named entity schema with implied entity types and span boundary ambiguity creates combinatorial disagreement that simple confidence thresholds cannot resolve. Standard active learning loops that depend on iterative model retraining are impractical at NGO scale. These gaps define the central problem this dissertation addresses: how to improve annotation quality on a real, single-annotator, specialist-schema dataset without access to reliable ground truth.

1.2 Research Questions

Four research questions structure the investigation:

1. To what extent do existing LLM-based annotation frameworks (Active-Prompt, LLMaAA, and Reflexion) improve annotation quality on a single-annotator MSHT dataset with a high-cardinality NER schema?

2. What structural properties of the dataset and schema drive persistent inter-sample disagreement in NER tasks, and why do these properties resist resolution by iterative self-critique?

3. Can a synthesis-based approach that treats model disagreement as evidence to be resolved rather than noise to be filtered produce internally consistent annotations without access to an external gold standard?

4. What are the practical deployment implications of a synthesis pipeline for an NGO operating under real budget and infrastructure constraints?

1.3 Aims and Objectives

The primary aim of this project is to improve the quality and consistency of crowdsourced annotations in a real-world MSHT dataset by evaluating established LLM-based annotation frameworks and, where they fall short, designing and implementing novel alternatives capable of handling the specific characteristics of the data. The work is conducted in partnership with Stop the Traffik and covers both English and Spanish data across NER and NLC tasks.

The specific objectives are:

1. To conduct an exploratory data analysis of the Stop The Traffik annotation dataset, examining annotator behaviour, label distributions, agreement patterns, and confidence levels across the dataset.

2. To assess the reliability of the original human annotations and determine whether inter-annotator agreement metrics are viable as a quality measure given the structure of the dataset.

3. To implement three established LLM-based annotation frameworks, Active-Prompt, LLMaAA, and Reflexion, as LangGraph pipelines and evaluate their performance on NER and NLC tasks across both languages.

4. To diagnose the specific characteristics of NER in this domain that make it structurally harder to annotate consistently than NLC, and to identify the limitations of each baseline framework in addressing those challenges.

5. To design and implement Active-Synthesis, a two-phase framework that combines uncertainty-driven sampling with LLM-based conflict resolution to produce stable, enriched annotations without treating human labels as a reference ceiling.

6. To evaluate Active-Synthesis against the three baseline frameworks and measure how much it improves annotation confidence, resolution rate, and coverage across the full dataset.

1.4 Methodology

This project follows four phases of iterative, empirical research: dataset characterisation, baseline framework evaluation, failure analysis, and novel framework development. Each phase shapes the next.

Phase 1: Exploratory Data Analysis: The Stop The Traffik annotation dataset was first examined in its raw form. A confidence scoring formula (based on annotator count, inter-annotator agreement, and a lead-time penalty for sub-10-second annotations) produced HIGH, MEDIUM, and LOW confidence tiers. As detailed in Chapter 3, the results showed the dataset was not inherently reliable, which led to the LLM-based enhancement approach.

Implementing and Evaluating Baseline Frameworks: Three frameworks (Active-Prompt, LLMaAA, and Reflexion) were implemented as LangGraph pipelines sharing the same infrastructure, so results could be compared directly. Each was evaluated

on 20 tasks per dataset (80 in total) across English and Spanish NER and NLC, enough to reveal behavioural patterns without full-dataset cost (Chapter 4).
Analysis: All frameworks performed reliably on NLC but struggled with NER. Active-Prompt detected uncertainty but produced no resolved outputs. LLMaAA failed on NER (0% confidence upgrades) because the 58-type schema makes exact span agreement effectively impossible. Reflexion achieved partial verification, with most English NER tasks failing to converge. These results pointed to specific issues (open-ended label spaces, span inconsistency, and implied entity types) that informed the design of Active-Synthesis.
Phase 4: Novel Framework Development and Evaluation: Active-Synthesis addresses structural disagreement by synthesising multiple conflicting outputs into a single justified annotation, rather than relying on iterative retries. As detailed in Chapters 4 and 5, it was piloted on 30 tasks per dataset and then run on the full dataset (215 English NER, 217 English NLC, and 1,120 tasks each for Spanish NER and NLC), which allowed comprehensive comparison across all four frameworks.

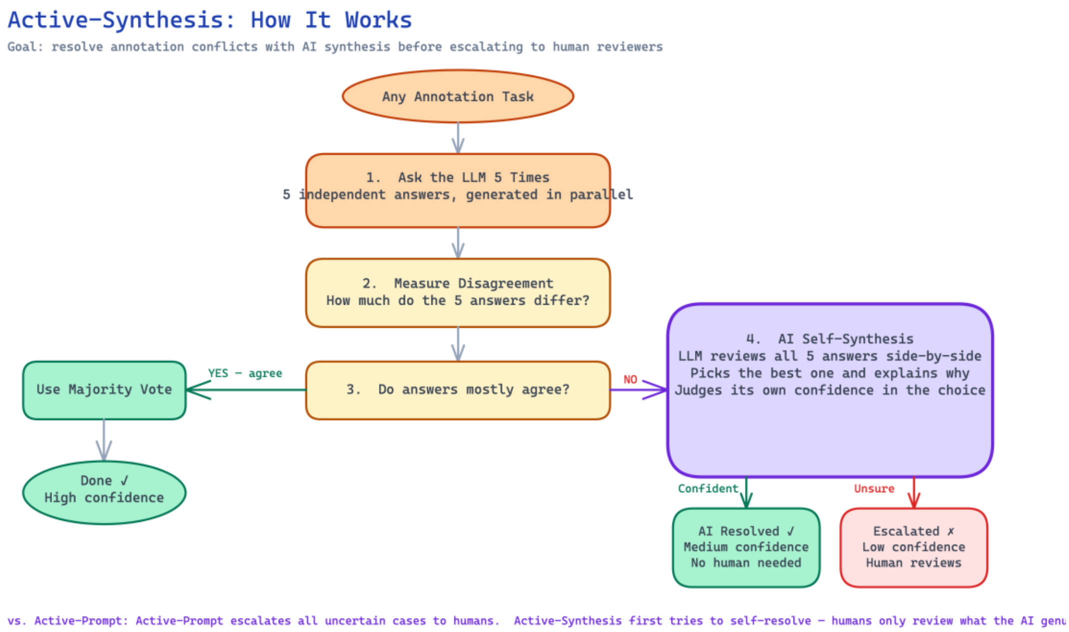

Figure 1.1: *Active-Synthesis: High Level Diagram*

1.5 Tools and Technologies

- **Python:** Primary programming language used throughout the project, chosen for its NLP and data analysis ecosystem and native support for asynchronous execution. All pipelines, shared infrastructure, and exploratory analysis were written in Python.

- **LangGraph**: Built on LangChain, orchestrates stateful, multi-step LLM workflows. Each annotation framework is implemented as a graph of processing nodes covering sampling, scoring, and synthesis, connected via conditional logic. This structure keeps the pipelines modular and straightforward to modify.

- **OpenAI API (GPT-4o)**: Handles all annotation tasks, chosen for its performance on structured output generation and multilingual text. All calls run at temperature 0.7 and are managed asynchronously with concurrency control and retry logic.

- **Pydantic:** Validates LLM outputs against predefined schemas, preventing invalid annotations from entering the pipelines.
- **Sentence Transformers:** The all-MiniLM-L6-v2 model generates embeddings for semantic similarity retrieval in the LLMaAA framework's few-shot prompting step.
- **Jupyter Notebook:** Used for exploratory data analysis.

1.6 Dissertation Structure

The remainder of this dissertation is structured as follows. Chapter 2 reviews the literature on annotation quality, active learning, LLM-based annotation, and NLP in the MSHT domain. Chapter 3 describes the dataset and presents the exploratory data analysis. Chapter 4 describes the design and implementation of all four pipelines. Chapter 5 reports results across all frameworks and datasets. Chapter 6 discusses findings, situates Active-Synthesis in the literature, and addresses limitations and ethics. Chapter 7 concludes with recommendations and future directions.

Chapter 2: Literature Review

2.1 Introduction

This chapter works through the research that shaped the methodology. It covers annotation quality and the limits of inter-annotator agreement, then moves to active learning, LLMs as annotators, and three specific frameworks (Active-Prompt, LLMaAA, and Reflexion) that this project implements and evaluates. It also examines CrowdAgent, a multi-agent system that is the most complete solution in the literature but could not be pursued here for practical reasons. The chapter ends by identifying the gap that motivated Active-Synthesis.

2.2 Annotation quality and ground truth challenges

Label quality is a basic precondition for supervised NLP, but it is rarely clean in practice. Alhazmi and Alsumari (2021) find that annotation errors degrade model performance in proportion to the error rate, though they also show that quantity gains from automated labelling can partially compensate, a trade-off that matters most when mislabelled examples carry real consequences. The problem compounds when annotators disagree. Sylolypavan et al. (2023) track how inter-annotator inconsistency in clinical AI pipelines flows directly into model uncertainty downstream, reducing reliability in ways that are difficult to detect without careful auditing. In MSHT detection, where labelling demands specialist knowledge and errors can distort the intelligence drawn from trafficking data, these concerns are not marginal.

In multi-annotator settings, the standard assumption that annotators are interchangeable rarely holds. Rodrigues et al. (2013) model annotator-specific reliability probabilistically, allowing conflicting labels to be weighted rather than simply majority-voted. Li et al. (2019) push this further: rather than assigning each annotator a single global reliability score, they model reliability at the instance level, since the same annotator can be accurate on some examples and unreliable on others depending on the specifics of the text. Dong et al. (2021) address label noise through co-training, using two classifiers trained on different feature views to iteratively correct each other's errors. Su et al. (2025) take a different angle with confident learning, which identifies mislabelled instances under the class-conditional noise assumption and handles non-uniform noise that most correction algorithms ignore.

What these papers share is the premise that annotation quality cannot be treated as a fixed input; it must be modelled. This dissertation adopts the same premise. When the Stop the Traffik dataset was examined, nearly all tasks had been labelled by a single annotator, which meant the standard apparatus of inter-annotator agreement (Cohen's Kappa, Fleiss' Kappa, percentage agreement) could not be applied. The confidence scoring approach developed in Chapter 3, which weights annotator count, agreement, and completion time into a three-tier quality classification, is a practical response to this constraint.

2.3 Active learning for efficient annotation

The core idea of active learning is that not all examples are equally worth labelling. If a model can identify which unlabelled instances would be most informative, annotation budgets can be stretched further and directed more precisely. Wang et al. (2021) review this across crowdsourced settings and confirm the general principle, but note an important caveat: naive active learning in noisy annotation environments can make things worse, amplifying label errors if annotator reliability is not accounted for. Wang et al. (2023) respond to exactly this, building label quality directly into the instance selection criterion so the framework avoids querying unreliable sources.

Uncertainty sampling is the most common selection heuristic: pick the examples the current model is least confident about. Sener and Savarese (2018) approach this geometrically through their Core-Set method, which selects examples that maximise coverage of the feature space and works well for deep classifiers without relying on calibrated probabilities. Ash et al. (2020) combine both considerations in BADGE (Batch Active learning by Diverse Gradient Embeddings), using gradient embeddings to capture uncertainty and clustering to ensure diversity within each batch. This avoids the redundancy problem that pure uncertainty sampling often produces, where a model queries many near-identical examples clustered around the same decision boundary. These two requirements, uncertainty and diversity, map directly onto the disagreement counting and Shannon entropy metrics used in this dissertation to score annotation confidence across parallel LLM runs.

Bernhardt et al. (2022) apply active learning logic to a different problem: rather than selecting new instances to annotate, they use it to find existing annotations that are likely wrong and correct them. In a setting where expert re-annotation is expensive, cleaning existing data can be more productive than acquiring more of it. This applies to the Stop the Traffik dataset, where the problem is not a shortage of labelled data but a shortage of reliably labelled data.

2.4 Large language models as annotators

The use of LLMs for annotation has moved from a curiosity to a practical option relatively quickly. Gilardi et al. (2023) compared ChatGPT directly against Mechanical Turk workers on text classification tasks and found ChatGPT achieved 91 to 97 percent inter-annotator agreement compared to 56 percent for crowd workers, at roughly a thirtieth of the cost. Results like these prompted a genuine reassessment of whether crowdsourcing should remain the default for NLP annotation. Bansal and Sharma (2023) tested a more systematic version through their EAGLE framework, showing that LLM-generated labels improve downstream generalisation when used to augment limited human-labelled data. Tan et al. (2024), surveying the broader space of LLM annotation methods, identify three consistent weaknesses across zero-shot, few-shot, and chain-of-thought approaches: consistency across runs, calibration of confidence, and generalisation to specialist domains.

That last weakness is the one that matters most here. Lu and Smith (2025) make the problem concrete: feeding LLM annotations directly into a BERT classifier can hurt performance compared to carefully curated human labels. LLM annotations need filtering, not blind acceptance. This is what motivates the confidence-tiered approach in this dissertation, where each annotation passes through uncertainty scoring before a final label is assigned. Meguellati et al. (2025) show a related result in harmful content detection: LLM-based data augmentation improves classifier robustness even in low-resource conditions where minority-class examples are scarce. MSHT detection sits in similar territory, with rare event types and limited labelled data, and the same logic applies.

A further consideration is the conservative tendency of human annotators, documented in the annotation report produced during the EDA phase. Individual task analysis showed that human labellers regularly captured only the most salient entities in an article (typically victim ages) while missing trafficker names, court details, location references, and recruitment methods present in the same text. This means the human labels function as a lower bound on what the data contains, not an authoritative ceiling. LLM enrichment, rather than LLM reproduction of incomplete labels, is therefore the appropriate goal.

2.5 Active learning integrated with large language models

Combining active learning with LLM annotation tries to capture the cost advantages of both. Zhang et al. (2023) do this directly in LLMaAA, placing an LLM annotator inside an active learning loop so the system queries only the instances where annotation is most likely to be informative. On NER and text classification benchmarks, LLMaAA approaches fully supervised performance while annotating far fewer examples. This is one of the three frameworks evaluated in this dissertation, though the implementation here departs substantially from the original design. The original LLMaAA relies on a BERT-based task model that retrains iteratively as annotations accumulate, using the model's uncertainty to drive instance selection across multiple rounds. In this project's setting, no model training infrastructure was available, so the LLMaAA pipeline is adapted to a single-pass architecture: semantic similarity retrieval via sentence embeddings replaces the iterative acquisition loop, and confidence is assessed in one step rather than updated across rounds. This implementation is more accurately described as a LLMaAA-inspired baseline than a faithful reproduction of the original, and any characterisation of the framework's limitations in Chapter 5 should be read in that light.

The choice of which examples to query matters as much as who does the annotating. Diao et al. (2023) address a known vulnerability in chain-of-thought prompting: the few-shot demonstrations in the prompt have an outsized effect on output quality and selecting them by hand does not scale. Active-Prompt fixes this by selecting demonstrations based on model uncertainty across multiple sampled outputs, specifically using disagreement counting and output entropy. These are the same metrics used in this dissertation to score annotation confidence, so Active-Prompt functions both as a source framework and as a direct methodological precedent for the uncertainty scoring in Active-Synthesis. Xiao et al. (2023) take the logic further in FreeAL, where an LLM and a small classifier iteratively refine labels without any human annotation at all, reaching near-supervised performance on classification tasks. This dissertation retains human oversight for low-confidence cases, but the iterative refinement structure FreeAL demonstrates informs the overall architecture.

2.6 Iterative refinement and verbal self-correction

A persistent limitation of single-pass LLM annotation is that there is no mechanism to catch and recover from errors. Shinn et al. (2023) address this with Reflexion, which enables an agent to reflect verbally on its own failed outputs and revise them across multiple attempts. Rather than retraining the model, Reflexion uses the model's own language to generate corrective feedback, stored in a sliding-window memory that informs subsequent attempts. On the HumanEval programming benchmark, Reflexion reaches a 91 percent pass rate; GPT-4 in a single pass achieves 80 percent. That gap points to something genuine: iterative verbal correction catches errors that one-shot generation leaves behind. Reflexion is the third baseline framework evaluated in this dissertation, adapted to drive the revision loop applied to low-confidence annotations.

Madaan et al. (2023) reach similar conclusions through SELF-REFINE, where an LLM critiques its own output and rewrites it without any external training signal or human feedback. Across seven generation tasks including dialogue, code, and sentiment transfer, SELF-REFINE outperforms single-pass generation in every condition. Read alongside Reflexion, the pattern is consistent: multi-step correction outperforms one-shot labelling even when the same model does both the generating and the critiquing.

However, the results in Chapter 5 expose an important limitation of this approach in the NER setting. When annotation disagreement is structural, arising from a large entity type schema and free-form span boundaries rather than genuine ambiguity, iterative self-reflection struggles to converge. The model repeats similar errors across trials not because it lacks insight into what went wrong, but because the underlying source of disagreement is the label space itself rather than a correctable reasoning failure. This distinction motivates Active-Synthesis.

2.7 Multi-Agent and synthesis frameworks

More recent work has explored whether distributing annotation across multiple interacting agents can overcome the limitations of single-agent iterative approaches. Du et al. (2023) demonstrate in their multi-agent debate framework that having multiple LLM instances independently reason and then debate their conclusions improves factual accuracy and reduces overconfidence compared to a single model, which aligns with the intuition behind sampling multiple independent annotations per task. Yao et al. (2024) extend this reasoning to complex problem-solving through Graph of Thoughts, where LLM outputs are organised as a graph rather than a linear chain, allowing solutions to be aggregated and refined across multiple reasoning paths.

The most complete integration of these ideas in an annotation-specific context is CrowdAgent (Qin et al., 2025), which proposes a multi-agent managed annotation system combining a small language model (SLM) and an LLM in a coordinated pipeline. The SLM is trained on a seed labelled set using noise-robust Gaussian Mixture Model (GMM)-based loss filtering to identify a clean coreset. It then scores all unlabelled examples by confidence and routes only the most uncertain cases to the LLM, which annotates them using few-shot demonstrations drawn from the coreset. The system achieves strong performance by ensuring the expensive LLM is applied only where the cheap SLM is genuinely uncertain, and by grounding LLM calls in high-quality representative examples.

CrowdAgent is the most theoretically well-motivated approach in the literature for the annotation quality problem this dissertation addresses. However, it could not be implemented in this project's setting for a practical reason: it requires training a BERT-based SLM on a clean seed set, which demands GPU compute resources and a sufficiently large validated labelled set to bootstrap reliable noise-robust training. Neither was available. The Stop The Traffik dataset had fewer than 2 percent of tasks at HIGH confidence, making a viable seed set difficult to construct, and the project operated within API-only constraints without access to model training infrastructure. CrowdAgent is therefore treated here as an upper bound for what a fully resourced annotation pipeline might achieve, and its core insight, routing uncertain cases to a more capable model while accepting easy cases from a cheaper one, is partially approximated in Active-Synthesis's tiered path architecture.

2.8 Named entity recognition in specialised domains

NER is harder than text classification in a specific way. Classification asks the model to pick a label for a whole span of text. NER asks it to find the span boundaries and assign a type simultaneously, two decisions that multiply in error. Transformer models handle this well in general-domain settings with small tagsets. With large, fine-grained schemas in specialist domains, they struggle. Bogdanova et al. (2025) show this in the biomedical domain, where open-schema NER requires models specifically adapted to domain vocabulary and annotation conventions.

The Stop the Traffik NER schema covers 58 entity types, including Trafficker Name, Victim Age, Victim Age (Implied Entity), Court Tribunal, Location Arrest, and Business of Interest. It is both large and unusual. Crucially, it includes a class of implied entity types that require inferential reasoning rather than surface extraction. A label like Victim Gender (Implied Entity) does not correspond to a text span that says "female" or "male"; it requires the annotator to infer gender from context, pronouns, or related mentions. This is a different cognitive operation from identifying a named location or a quoted age, and it introduces a category of ambiguity that consistency-based metrics cannot resolve.

This distinction between extractive and inferential entity types proved central to understanding why NER was structurally harder than NLC in this project. Disagreement between LLM runs on NLC tasks was low because the bounded label space of six classifiers with five to eight categories each naturally converges. Disagreement on NER tasks was high because the same article text can support multiple defensible span-boundary decisions and multiple valid type assignments for implied entities. No amount of iterative self-reflection changes this if the source of disagreement is the nature of the label rather than an error in reasoning.

2.9 Synthesis and research gap

These findings are reasonably consistent across the literature. Annotation quality varies by instance and by annotator, and frameworks that ignore this pay for it downstream (Rodrigues et al., 2013; Li et al., 2019). LLMs can annotate at a fraction of the cost of crowd workers, but blind acceptance of their outputs degrades model performance (Lu and Smith, 2025), so uncertainty-based filtering is necessary rather than optional. Active learning selection works best when it balances uncertainty against diversity (Ash et al., 2020; Sener and Savarese, 2018) and can be applied to correcting existing labels as much as acquiring new ones (Bernhardt et al., 2022). Iterative self-correction improves on single-pass annotation but struggles when disagreement is structural rather than incidental (Shinn et al., 2023; Madaan et al., 2023).

None of this work, however, fully addresses the combination of challenges present in this project's setting: a specialist multilingual dataset with a large, fine-grained entity schema that includes implied entity types, an annotation base that is predominantly single-annotator and of low confidence, and constraints that preclude model training. Most LLM annotation frameworks are benchmarked on general-domain English text with clean evaluation conditions and access to ground truth. The approach proposed in this dissertation, Active-Synthesis, addresses these conditions by combining uncertainty sampling (drawn from Active-Prompt) with a conflict resolution step in which the LLM is presented with all its own conflicting outputs simultaneously and asked to synthesise a single justified final annotation. This differs from Reflexion-style iterative correction, which attempts to converge toward a human label through sequential self-critique, and from LLMaAA-style few-shot retrieval, which relies on a pool of reliable demonstrations that was not available in this setting. Active-Synthesis is designed specifically for the case where disagreement is high, human labels are incomplete, and the goal is not replication of existing annotations but production of richer, more consistent labels that can support downstream model training.

Table 2.1: Comparison of related work in LLM-based annotation and quality modelling

Author(s) & Year	Method / Approach	Dataset Used	Key Contribution	Limitations relative to this work
Gilardi et al. (2023)	Zero-shot GPT-3.5 annotation compared directly against Mechanical Turk crowd workers	Political tweets and news articles (English)	ChatGPT achieves 91–97% inter-annotator agreement vs. 56% for crowd workers at ~1/30th the cost; shows LLMs are viable annotation alternatives	Evaluated on simple text classification with small, clean label sets; does not address high-cardinality schemas, NER, or multilingual data

Zhang et al. (2023)	LLMaAA: active learning loop using an LLM annotator guided by a BERT task model that retrains iteratively	CoNLL-2003 (NER), SST-2, TREC (classification)	Approaches fully supervised performance while annotating far fewer examples, by combining LLM annotation with uncertainty-driven instance selection	Requires model training infrastructure and a pool of reliable seed annotations; evaluated on standard English benchmarks, not specialist schemas
Diao et al. (2023)	Active-Prompt: selects few-shot CoT demonstrations by measuring model uncertainty (disagreement + entropy) across multiple sampled outputs	Eight reasoning benchmarks (arithmetic, commonsense, symbolic)	Outperforms static CoT prompting by selecting demonstrations where model disagreement is highest; introduces disagreement and entropy as practical uncertainty signals	Designed for reasoning tasks with clean evaluation; not applied to annotation or NER; does not address structural disagreement in large label spaces
Shinn et al. (2023)	Reflexion: verbal self-critique loop where an LLM reflects on failed outputs and revises them across multiple trials using sliding-window memory	HumanEval (code), AlfWorld (embodied tasks), HotpotQA	Reaches 91% pass rate on HumanEval vs. 80% for GPT-4 single-pass; demonstrates that iterative verbal correction systematically outperforms one-shot generation	Convergence depends on the existence of a correctable reasoning error; fails when disagreement is structural rather than incidental, as shown in this study's NER results
Madaan et al. (2023)	SELF-REFINE: LLM critiques and rewrites its own output iteratively without external feedback or retraining	Seven generation tasks including code, dialogue, sentiment	Outperforms single-pass generation across all seven tasks; confirms multi-step correction as a general improvement strategy	Same structural limitation as Reflexion: cannot converge when the source of disagreement is the label space rather than a reasoning error

Bernhardt et al. (2022)	Active label cleaning: uses active learning to identify likely mislabelled examples in existing datasets and correct them	Chest X-ray pathology classification (CheXpert)	Shows that cleaning existing labels can be more productive than acquiring new ones under resource constraints	Requires a trained model and partial ground truth to identify suspect labels; not applicable where a clean seed set is unavailable
Rodrigues et al. (2013)	Probabilistic annotator reliability modelling: weights conflicting labels by annotator-specific reliability rather than majority voting	Sentiment analysis, text classification (crowdsourced)	Demonstrates that annotator reliability varies and that weighting labels by estimated reliability improves downstream model performance	Requires sufficient multi-annotator overlap to estimate reliability; inapplicable when 98%+ of tasks have a single annotator
Lu & Smith (2025)	Empirical comparison of BERT classifiers trained on LLM-generated labels vs. carefully curated human labels	Text classification benchmarks	Shows that feeding LLM annotations directly into a BERT classifier can hurt performance; uncertainty-based filtering of LLM outputs is necessary	Focuses on classification tasks with clean evaluation conditions; does not address the NER enrichment problem or incomplete human baselines
Qin et al. (2025)	CrowdAgent: multi-agent pipeline combining a noise-robust SLM (trained with GMM-based loss filtering) and an LLM for selective	Text classification (crowdsourced benchmarks)	Routes uncertain cases to LLM and easy cases to SLM; achieves strong performance by grounding LLM calls in a verified high-quality coreset	Requires GPU infrastructure for SLM training and a clean seed set; neither was available in this project's setting

	annotation routing			
Yazdani et al. (2025)	GLiNER-BioMed: compact bidirectional encoder trained to classify token spans against schema-defined entity descriptions at inference time	Biomedical NER benchmarks	Achieves state-of-the-art specialist NER performance at a fraction of the inference cost of large generative models	Requires labelled training data and a model training phase; most relevant as a downstream target once Active-Synthesis has generated a validated annotation pool

Chapter 3: Dataset Description and Exploratory Data Analysis

3.1 Dataset Description

The dataset was provided by Stop the Traffik and consists of news articles on Modern Slavery and Human Trafficking in English and Spanish. Annotations were collected through Label Studio, an open-source annotation platform. The raw data was exported from Label Studio as structured JSON files, one per task type and language combination, and is the primary input for all subsequent analysis and pipeline work.

Before any analysis was conducted, the dataset underwent a PII removal pass to anonymise personally identifiable information of the annotators. A PII mapping file recorded all substitutions, and the cleaned versions of all four files were used as the input for all pipelines. This step was necessary both for ethical reasons and to comply with the data sharing agreement with Stop the Traffik.

In total the dataset comprises 2,800 tasks, 2,875 annotations, 16 annotators, and 27,911 label instances. The Spanish batches are approximately five times larger than the English batches, and Spanish is the dominant workload. English tasks achieve 100% coverage, while Spanish tasks reach 95% (1,120 of 1,184 tasks annotated), with a small number having no submitted annotation.

The dataset is structured across four files:

Table 3.1: Summary of Dataset Files and Task Counts

File	Language	Task	Tasks
7-English_NER_Articles	English	NER	215
7-English_NLC_Articles	English	NLC	217
9-NER_Questions_Batch_9_Spanish	Spanish	NER	1,184
9-NLC_Questions_Batch_9_Spanish	Spanish	NLC	1,184

3.2 Annotation schema

Each task is annotated using one of two structured schemas, depending on its type.

Named Entity Recognition (NER)
Annotators identify text spans and assign them to one of 58 predefined entity types covering trafficking-relevant information (e.g., Trafficker Name, Victim Age, Location Reference, Court Tribunal, Investigating Agency, Business of Interest, Victim Count, Trafficker Gender). The schema also includes implied entity types (e.g., Victim Age, Trafficker Gender, Business of Interest) that require inference from context rather than explicit mentions. For example, “the girls” implies victim gender, and “massage parlour” implies a business of interest. This distinction between extractive and inferential annotation contributes to NER being structurally more challenging than NLC.

Natural Language Classification (NLC)
Annotators assign category labels across six classifiers: incident type, coercion method, recruitment tactic, transport method, trafficking type, and trafficking sub-type. Each classifier has a bounded set of categories, allows multiple labels per article, and includes an ABSTAIN option for ambiguity. Compared to NER's open-ended span selection and 58 entity types, NLC's fixed label space results in a more constrained output structure.

3.3 Exploratory Data Analysis (EDA)

3.3.1 Dataset overview

Figure 3.1: *Dataset composition overview showing task and annotation counts*

The four files are structurally consistent: annotation counts closely match task counts across all of them, and completion rates are high. English files have between 312 and 314 annotations across 215 to 217 tasks, the difference comes from a small number of tasks that received multiple annotations. Spanish files have 1,123 to 1,126 annotations across 1,184 tasks, with 95% coverage. Draft annotations (incomplete submissions) were present in small numbers across all files and were excluded from quality analysis.

3.3.2 Annotator workload

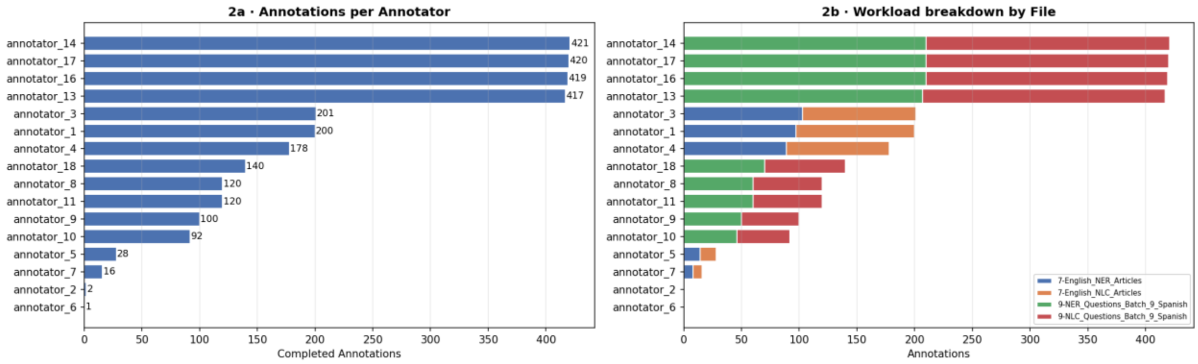

Figure 3.2: *Annotator workload distribution across all contributors*

Sixteen annotators contributed to the dataset, but workload was highly uneven. Four annotators (13, 14, 16, and 17) each completed between 417 and 421 annotations and worked exclusively on the Spanish batches. Together they account for approximately 60% of all annotations. Three annotators (1, 3, and 4) each completed between 178 and 201 annotations and worked exclusively on the English batches. The remaining nine annotators contributed between 1 and 140 annotations each, with two annotators (2 and 6) contributing only 2 and 1 annotation respectively.

This concentration of workload creates a significant risk for the downstream quality of the Spanish dataset. If the four dominant Spanish annotators share systematic biases (common label interpretations, consistent omissions, or a similar threshold for ABSTAIN), those patterns will be replicated across the entire Spanish corpus without any counterbalancing annotator to surface disagreement. The English dataset is more evenly distributed across three primary contributors, which gives more opportunity for natural disagreement to surface, though still with limited overlap.

3.3.3 Inter-annotator agreement and confidence classification

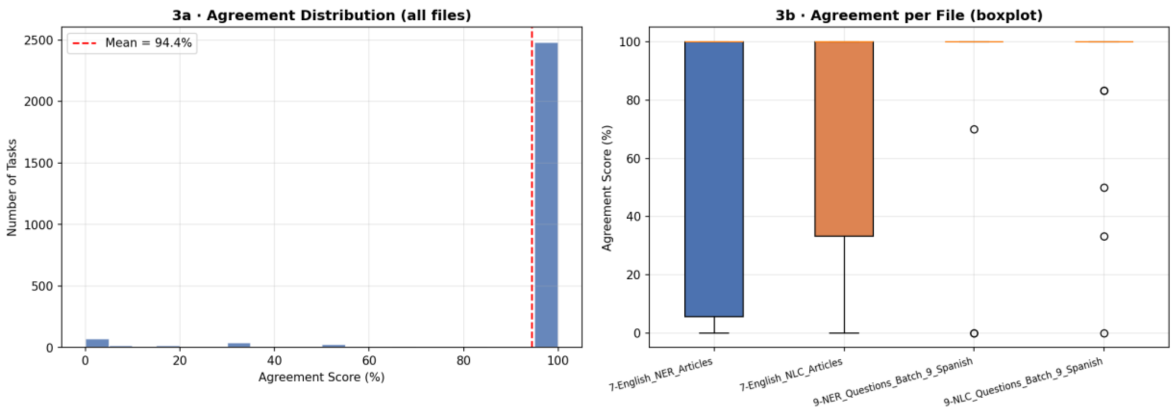

Figure 3.3: *Inter-annotator agreement distribution across multi-annotator tasks*

The headline inter-annotator agreement figure reported by Label Studio (a mean of 94.4% and a median of 100%) is highly misleading. It is computed across all tasks, including the 92.4% that had only a single annotator. For single-annotator tasks, Label Studio records agreement as 100% by default, since there is no second annotation to compare against. This inflates the aggregate figure dramatically.

When agreement is computed only on the 203 tasks (7.6%) that had two or more annotators, the picture is substantially different. Mean agreement on these tasks was 26.1%, with a median of 16.7%. Only 4.9% of multi-annotator tasks reached full agreement, and 74.9% had agreement below 50%. These figures make inter-annotator agreement unviable as a quality metric, not because agreement is globally poor, but because the dataset was not designed for it. The annotation pipeline did not routinely assign multiple annotators to the same task, and the overlap that does exist is too sparse and inconsistent to support reliable agreement measurement.

In response, I developed a three-tier confidence classification that can be applied to every task regardless of annotator count. The formula weights three factors: annotator count (up to 0.20 of the score), inter-annotator agreement where available (up to 0.70), and a lead-time quality penalty that down-weights annotations completed in under ten seconds (up to 0.10). Tasks were classified as HIGH (two or more annotators, 100% agreement), MEDIUM (two or more annotators, 50-99% agreement), or LOW (single annotator, or agreement below 50%).

The results of this classification are summarised below and shown in full in Table 3.2.

Table 3.2: Confidence Classification Results by Dataset File

File	HIGH	MEDIUM	LOW	Total
English NER	4 (2%)	2 (1%)	209 (97%)	215
English NLC	5 (2%)	35 (16%)	177 (82%)	217
Spanish NER	0 (0%)	1 (0%)	1,183 (100%)	1,184
Spanish NLC	1 (0%)	3 (0%)	1,180 (100%)	1,184
Total	**10 (0.4%)**	**41 (1.5%)**	**2,749 (98.2%)**	**2,800**

Across the full dataset, 98.2% of tasks were classified as LOW confidence. The Spanish dataset has effectively zero multi-annotator coverage: one HIGH task across 2,368 Spanish tasks combined. These findings drove the design of this project. The existing annotations could not be treated as reliable ground truth, and an alternative approach to annotation quality was needed.

3.3.4 Label distributions — NER

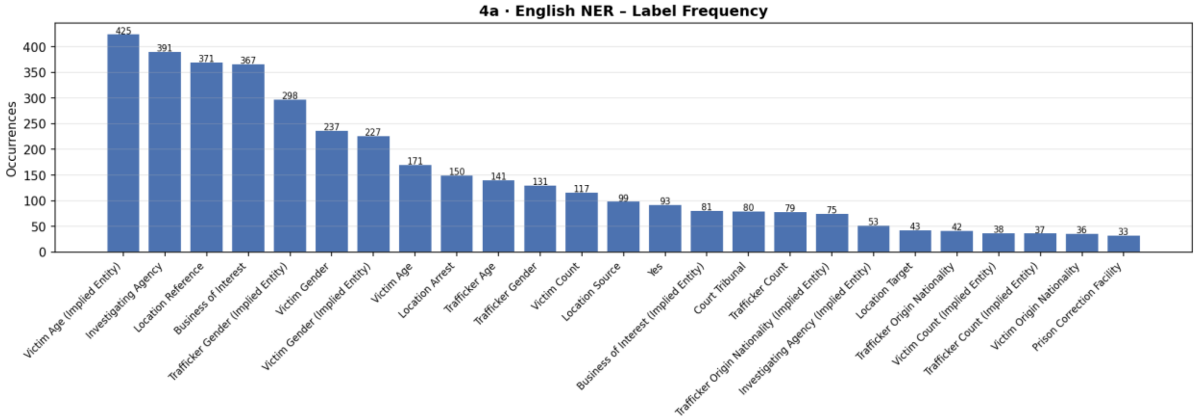

Figure 3.4: *NER entity label frequency distribution (English dataset)*

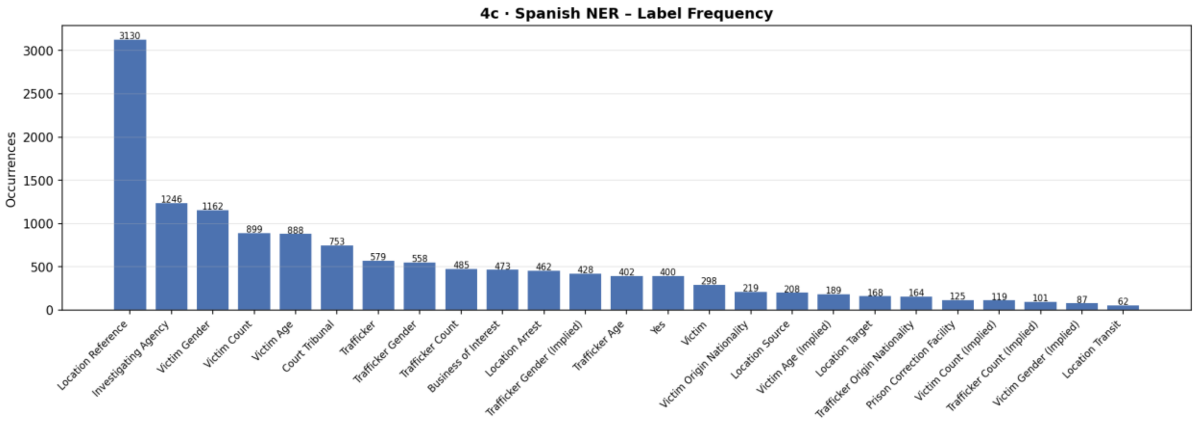

Figure 3.5: *NER entity label frequency distribution (Spanish dataset)*

English NER annotations span 37 unique entity types across 3,958 total spans. The most frequent labels are Victim Age (Implied Entity) (425 instances), Investigating Agency (391), and Location Reference (371). The prominence of implied entity types in the top labels is notable: three of the fifteen most frequent labels are implied variants. This confirms that inferential annotation was a regular and expected part of the task, not an edge case.

Spanish NER covers 58 unique labels across 13,920 total spans. The distribution is heavily skewed: Location Reference alone accounts for 3,130 spans, more than double the next most frequent label, Investigating Agency (1,246). Labels such as Court Tribunal (753) and Trafficker Count (485) appear frequently in Spanish but were less prominent in English. Whether this reflects the types of articles in each language subset, or different annotation habits across the Spanish and English annotator groups, is hard to separate from the data alone.

A long tail of rare labels is present in both languages. Several label types appear fewer than five times across the entire dataset. These low-frequency types will be the most challenging for any LLM-based annotation framework, as the model will have had limited in-context exposure to guide its decisions.

3.3.5 Label distributions — NLC

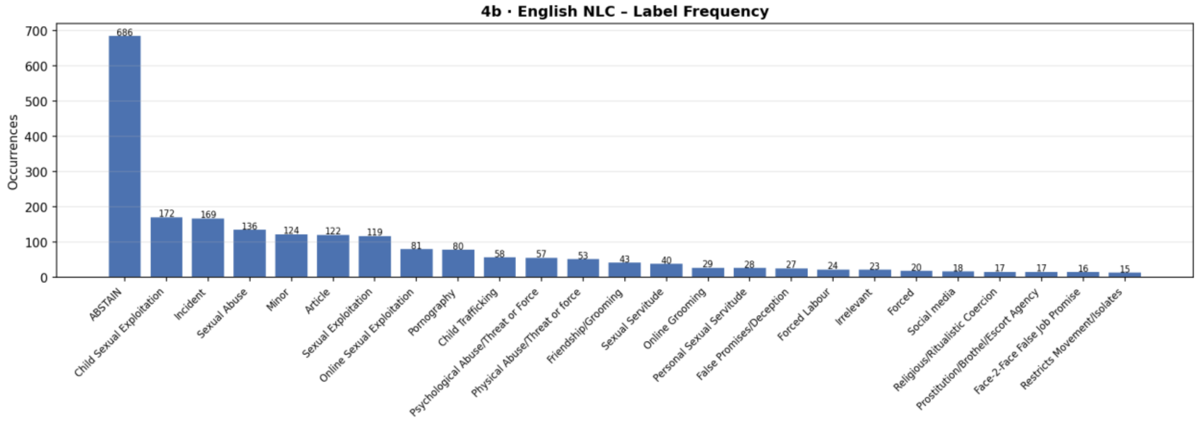

Figure 3.6: *NLC label frequency distribution (English dataset)*

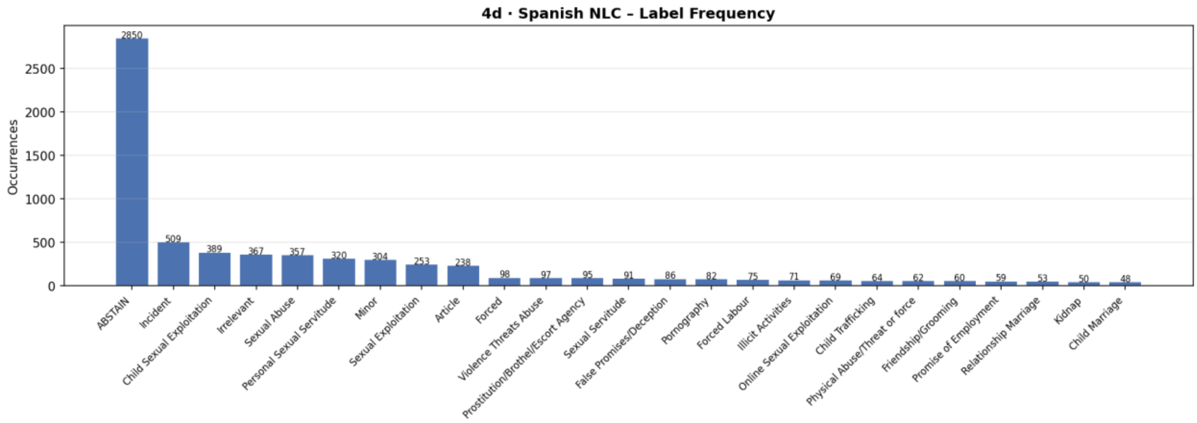

Figure 3.7: *NLC label frequency distribution (Spanish dataset)*

The NLC label distributions are dominated by ABSTAIN in both languages. In English NLC, ABSTAIN accounts for 686 instances, approximately four times the next most frequent label, Child Sexual Exploitation (172). In Spanish NLC, ABSTAIN accounts for 2,850 instances, more than five times Incident (509), the next most frequent.

This high ABSTAIN rate reflects the multi-label, multi-classifier structure of the NLC task. For any given article, many of the six classifiers will be inapplicable; a report about sexual exploitation may not involve any recruitment tactic worth labelling, and so the recruitment classifier receives an ABSTAIN. The high ABSTAIN volume is a structural feature of the schema rather than evidence of annotator uncertainty, though the two are difficult to distinguish without deeper per-task inspection.

3.3.6 Annotation speed and quality flags

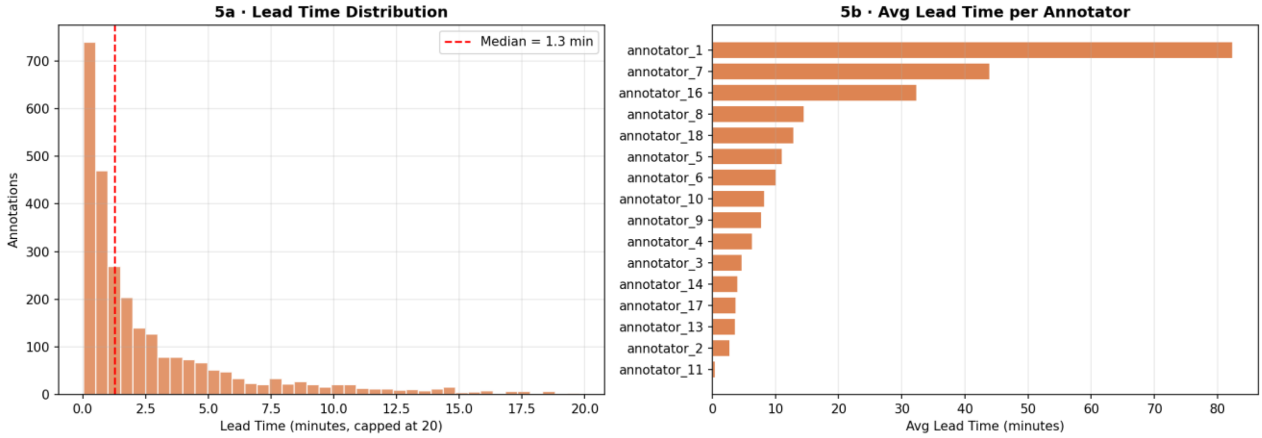

Figure 3.8: *Annotation speed distribution by annotator*

Annotation speed varied enormously across annotators. The global median lead time was 1.4 minutes per task, but the mean was 15.0 minutes, pulled upward by a small number of annotators who took much longer. Annotator 1 averaged 82.4 minutes per task across 200 annotations, and annotator 7 averaged 44 minutes across 16. At the other extreme, annotator 11 had a median lead time of 4.6 seconds and annotator 17 a median of 10.2 seconds.

A total of 969 annotations were flagged as suspiciously fast using two criteria: completion in under ten seconds (an absolute threshold), or completion in less than half the annotator's own median lead time (a relative threshold). Annotator 11 had 82 of their 120 annotations flagged, and annotator 4 had several tasks recorded at under two seconds, physically insufficient time to read a 350-word article, let alone annotate it. These annotations were assigned a lead-time quality penalty in the confidence scoring formula and may represent systematic rushing or platform interactions that auto-submitted responses.

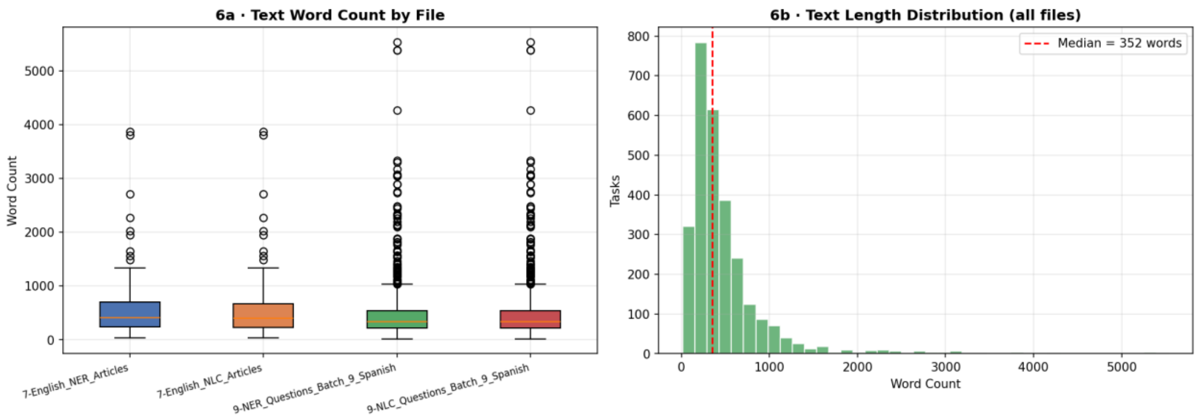

Figure 3.9: *Article text length distribution across the corpus*

Article lengths follow a right-skewed distribution. The median word count is 352 words and the mean is 486, with a range from 17 to 5,535 words. All four files look similar, which suggests the source article selection process was applied consistently regardless of language or task type. Long documents (those exceeding 1,000 words) are a small proportion of the corpus but matter for annotation quality: longer texts offer more potential entity mentions and create greater cognitive load. They also present a token budget problem for LLM-based annotation, since very long articles may approach or exceed prompt length limits.

3.3.8 Temporal patterns and review outcomes

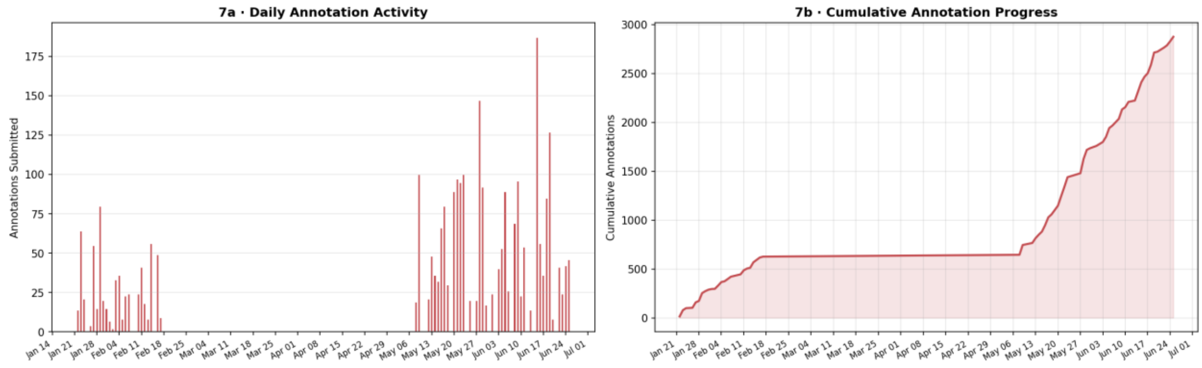

Figure 3.10: *Cumulative annotation activity over the annotation period*

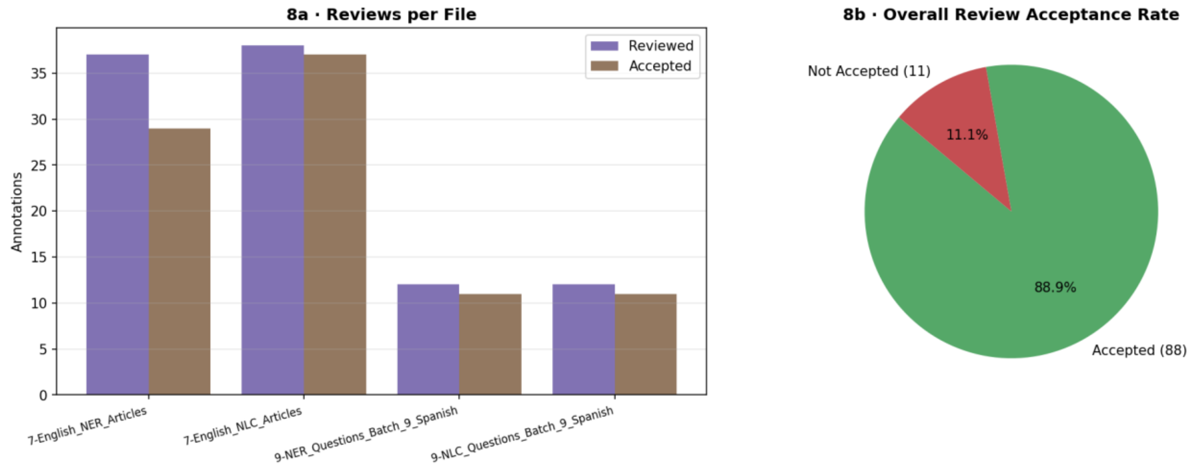

Figure 3.11: *Review outcome distribution across annotation submissions*

Annotation activity ran from 22 January to 25 June 2025, a window of 153 days. The cumulative progress curve shows two distinct phases: a slow early period through April in which approximately 600 annotations were completed, followed by a sharp acceleration in May and June, with a peak of 187 annotations on 15 June 2025. This temporal clustering likely reflects volunteer availability and the timing of annotation drives organised by Stop the Traffik, and it means that a large proportion of the corpus was annotated in a compressed period with fewer opportunities for organiser review.

Of the 2,875 annotations, only 99 were formally reviewed (3.4%). Of those, 88 were accepted (88.9% acceptance rate) and 11 were rejected. English NLC had the most reviews relative to its size; Spanish batches had the fewest. The very low review rate means that the formal review process provides almost no quality signal at the dataset level. Even the 11 rejected annotations represent less than 0.4% of the total corpus.

3.4 Key findings and implications

The EDA produced four findings that directly shaped the design of the annotation enhancement pipelines described in Chapter 4.

The most basic problem is single-annotator dominance. With 98.2% of tasks having only one annotation, inter-annotator agreement cannot serve as a quality measure. The confidence classification formula developed in this chapter, weighting annotator count, agreement, and lead-time quality, is a practical substitute, but it cannot compensate for the absence of multiple independent annotations on the same task.

The quality picture that follows from this is not ambiguous. 98.2% of tasks are classified as LOW confidence. The Spanish dataset is effectively unvalidated, with a single HIGH confidence task across 2,368 entries. Even the English dataset, which has more annotator overlap, has 82-97% of tasks at LOW confidence depending on the file.

Individual task inspection added a further complication. Annotators regularly labelled only the most salient entities in an article (typically victim ages) while missing court details, trafficker identities, location references, and recruitment context. This means the existing labels are a lower bound on what the data contains, not an authoritative record of it. LLM annotation enhancement should therefore aim for enrichment and consistency rather than reproduction.

Chapter 4: Methodology and framework design

4.1 Overview

This chapter describes the methodological approach taken to address the annotation quality problem identified in Chapter 3. The project evaluated four LLM-based annotation frameworks (Active-Prompt, LLMaAA, Reflexion, and Active-Synthesis) across two task types (Named Entity Recognition and Natural Language Classification) and two languages (English and Spanish). Each framework was implemented as a standalone pipeline using LangGraph, a graph-based orchestration library for large language models, and was applied to the same dataset of Label Studio exports.

The four pipelines share common infrastructure: a unified data schema, a shared prompt library, and consistent scoring modules. Each contributes a distinct annotation strategy. Active-Prompt performs uncertainty triage to identify which tasks most require human attention. LLMaAA re-annotates low-confidence tasks using few-shot examples retrieved from a pool of higher-confidence cases. Reflexion attempts to iteratively resolve disagreement between a model and a human label through verbal self-critique. Active-Synthesis, this project's own contribution, extends the uncertainty-sampling step with a conflict-resolution mechanism that synthesises all conflicting model outputs into a single justified annotation.

Each framework section covers the graph architecture, scoring logic, prompt design, and configuration parameters.

4.2 Shared infrastructure

4.2.1 Orchestration with LangGraph

All four pipelines are implemented using LangGraph, a library that models multi-step LLM workflows as directed graphs. Each pipeline defines a typed state object (TypedDict), a set of node functions that transform that state, and conditional edge functions that determine which node to invoke next based on the current state. This design has several engineering advantages relevant to annotation pipelines: the full decision history is captured in a shared state object, conditional branching is explicit and testable, and each node can be developed and unit-tested in isolation.

The state objects across the four pipelines follow a common pattern. At minimum, each state carries the task identifier, the article text, the task type (NER or NLC), the language, and the human annotation. Additional fields accumulate as the task moves through the graph: uncertainty scores, triage tier, LLM outputs, agreement scores, and an audit trail that logs every decision made at every node. I designed this audit trail to support post-hoc analysis of pipeline behaviour without requiring re-execution.

4.2.2 Annotation schema

The NER annotation schema comprises 58 entity types organised by data type. Numeric types capture victim and trafficker ages as exact text spans; gender types extract gendered words or kinship terms verbatim; location types differentiate between arrest locations, transit locations, and general geographic references; and facility types identify institution names. Implied entity types cover cases where an entity is not explicitly named but is strongly implied by a phrase in the text, with the span required to be the exact evidential phrase rather than an inferred entity name.

The NLC schema provides six classifiers: incident, coercion, recruitment, transport, trafficking type, and trafficking type sub-classifier. Each classifier has a bounded set of valid categories, with ABSTAIN as a valid label when no category applies. Stop The Traffik provided the schema as part of the original annotation project; it was not modified for this study.

4.2.3 Shared prompts

A shared prompt library is used by all four pipelines. For NER tasks, the system prompt instructs the model to extract entity spans verbatim (no paraphrasing or normalisation), to select the most specific applicable entity type, and to avoid duplicating span-type pairs. Specific rules address common edge cases identified during exploratory analysis, including the handling of age mentions adjacent to prison sentences, gender inference from kinship terms, and the conditions under which implied entities should be extracted. For NLC tasks, the system prompt defines each classifier and its valid categories and instructs the model to resolve the most salient classifier first.

A Chain-of-Thought (CoT) variant of the NER prompt was also developed, which requests a rationale field alongside the span output. Active-Prompt's annotate_node uses this CoT prompt for top-priority tasks, giving human reviewers evidence to evaluate the model's reasoning.

4.2.4 Reliability engineering

All LLM calls are issued with GPT-4o at a temperature of 0.7. For the sampling calls that generate five parallel outputs per task, this non-zero temperature is intentional: it produces the inter-sample variability required to generate a meaningful uncertainty signal from disagreement and entropy scores. For single-call annotation, synthesis, and re-annotation steps, the same temperature setting is applied consistently across all pipelines. Output validation is handled through a Pydantic-style schema check: each LLM response is parsed from JSON and validated against the expected structure for the task type, with invalid outputs discarded and logged to the audit trail rather than silently failing. Parallel LLM calls use Python's asyncio concurrency primitives, with a configurable semaphore limiting simultaneous requests to avoid rate-limit errors. Transient API failures are handled through exponential backoff implemented with the tenacity library.

4.3 Active-Prompt: uncertainty-driven triage

4.3.1 Motivation and design

Active-Prompt, introduced by Diao et al. (2023), combines uncertainty quantification with Chain-of-Thought prompting. The original method selects the most uncertain training examples for human annotation and uses those annotations to build a CoT demonstration set. This project adapts the core uncertainty-sampling mechanism for the annotation validation context, applying it as a triage filter that identifies which tasks exhibit the highest model disagreement and therefore most require human attention.

4.3.2 Graph architecture

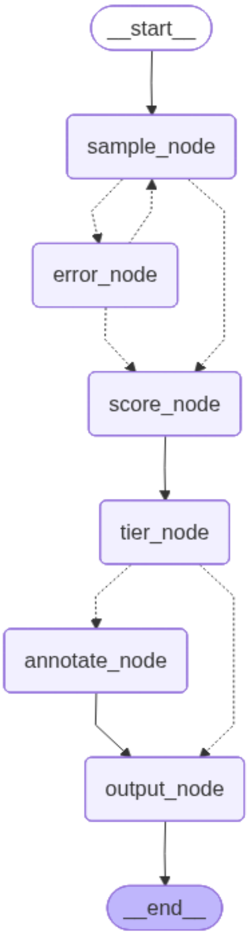

Figure 4.1: *Active-Prompt LangGraph pipeline architecture*

The Active-Prompt graph comprises six nodes. The sample_node runs five parallel LLM calls against the same article and validates each output against the task schema, discarding invalid responses and logging any discards to the audit trail. The score_node computes two uncertainty metrics from the validated samples: a disagreement count and Shannon entropy. For NER tasks, disagreement is defined as the number of span-type pairs that appear in fewer than half of the five samples; this measures how many entities the model failed to agree on across runs. For NLC tasks, disagreement counts the number of distinct incident classifier labels across samples. Shannon entropy is computed over the frequency distribution of span-type pairs (NER) or incident labels (NLC), as a continuous uncertainty signal that complements the discrete disagreement count.

The tier_node assigns the task to a triage tier using language-specific thresholds. For English tasks, a disagreement score exceeding three or an entropy score exceeding 1.5 places the task in the TOP tier. Cases falling within a moderate range on either metric are assigned to the MID tier. Tasks falling below both thresholds, where model agreement is high, are placed in the BOTTOM tier. Thresholds for

Spanish tasks are set higher because the model exhibits greater output variability on Spanish articles. I established these values empirically through inspection of the disagreement distributions produced during exploratory analysis.

TOP-tier tasks are routed to the annotate_node, which issues a single Chain-of-Thought call requesting a detailed rationale alongside the annotation. This rationale is intended to help a human reviewer understand which entities or categories the model identified and why. After the CoT call, or after scoring for MID and BOTTOM tasks, all tasks pass through the output_node, which maps the triage tier to a confidence label: BOTTOM to HIGH, MID to MEDIUM, and TOP to LOW. An error_node handles API failures with up to two retries before routing the task through the standard scoring path.

4.3.3 Scoring logic

Shannon entropy is computed as $H = -\Sigma\, p(x) \log_2 p(x)$, where $p(x)$ is the proportion of samples in which a given span-type pair (NER) or incident label (NLC) appeared. A value of zero indicates perfect agreement across all samples; higher values indicate greater distributional spread. For NER, the majority-vote annotation (the set of span-type pairs that appeared in at least half of the samples) is also retained in the task state for downstream use.

4.4 LLMaAA: few-shot re-annotation with KNN retrieval

4.4.1 Motivation and design

LLMaAA (LLM as an Annotation Advisor), proposed by Wang et al. (2023), uses a model to re-annotate low-confidence items with the assistance of few-shot demonstrations drawn from a pool of already-verified cases. A small number of carefully selected examples, retrieved based on semantic similarity to the target task, can substantially improve annotation quality without requiring model fine-tuning.

In this implementation, the demo pool is constructed from tasks classified as HIGH or MEDIUM confidence by the Active-Prompt triage step. Tasks flagged as LOW confidence (meaning the model exhibited high disagreement) are submitted for re-annotation with KNN-retrieved examples in their prompt context. The pipeline then evaluates the LLM's re-annotation against the original human label and assigns an upgraded confidence level based on the agreement score.

4.4.2 Graph architecture

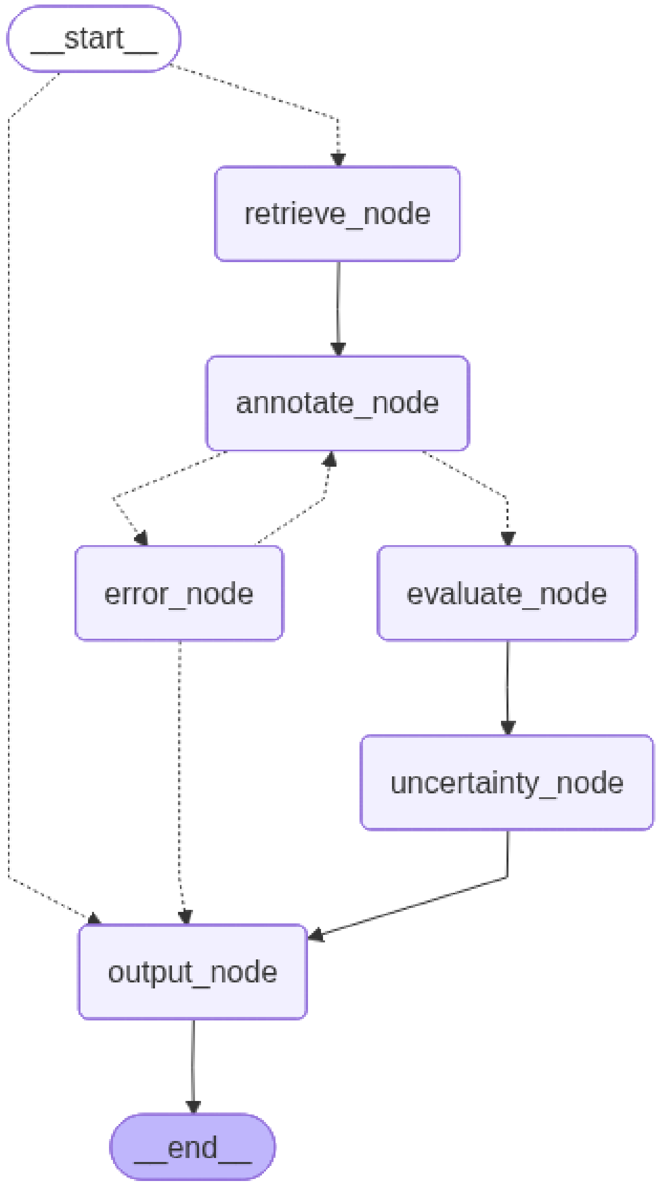

Figure 4.2: *LLMaAA LangGraph pipeline architecture with KNN retrieval*

The LLMaAA graph contains five nodes. Before any tasks are processed, a demo pool is built by embedding the article texts of all HIGH and MEDIUM confidence tasks using the paraphrase-multilingual-mpnet-base-v2 sentence transformer. I selected this multilingual model to support the Spanish articles in the dataset. Embeddings are computed in batch and stored in a numpy array for cosine similarity retrieval.

For each LOW-confidence task, the retrieve_node fetches the five nearest neighbours from the demo pool by cosine similarity, excluding exact matches. These demonstrations are injected into the prompt context alongside the task schema and the target article. The annotate_node then issues a single LLM call at temperature 0.7 to produce a re-annotation. The evaluate_node scores the re-annotation against the human label using span-level F1 for NER tasks and average per-classifier F1 for

NLC tasks, and maps the score to a new confidence level: a perfect match earns HIGH confidence, a partial match above 0.5 earns MEDIUM, and anything at or below 0.5 remains LOW. Finally, the uncertainty_node runs five additional LLM calls to estimate output stability, reporting the disagreement across those runs as an uncertainty score.

4.4.3 Limitations in NER re-annotation

The LLMaAA pipeline produced no confidence upgrades for NER tasks across either language. The root cause lies in the interaction between the scoring metric and the annotation schema. Span-level F1 requires an exact match on both the span text and the entity type for a pair to count as a true positive. With 58 entity types and free-form span boundaries, the probability of the model producing an annotation that overlaps exactly with the human label, even when the model's annotation is semantically reasonable, is very low. Two annotations can agree on the relevant entities while diverging on span boundary, capitalisation, or entity type assignment, producing an F1 of zero despite genuine agreement. This is a fundamental incompatibility between the F1-based upgrade criterion and the combinatorial label space of the NER schema, rather than a failure of the model itself. NLC tasks, with their bounded label sets and classifier structure, were not subject to this problem, and the pipeline produced meaningful upgrades across those tasks.

4.5 Reflexion: iterative self-critique

4.5.1 Motivation and design

Reflexion, introduced by Shinn et al. (2023), is a framework in which an LLM uses verbal self-critique to improve its outputs iteratively. After producing an initial annotation, the model evaluates its own output, identifies errors, and generates a natural language reflection. This reflection is injected into the prompt context of the next attempt, guiding the model toward a corrected annotation. The loop continues until the annotation reaches a sufficient agreement threshold or a maximum number of trials is exhausted.

In this implementation, Reflexion is applied as a validation step. Tasks that have already been classified as HIGH or MEDIUM confidence pass through the pipeline without entering the loop, as they are considered sufficiently reliable. Only LOW-confidence tasks, where initial model uncertainty was high, are subjected to the iterative annotation-evaluate-reflect cycle.

4.5.2 Graph architecture

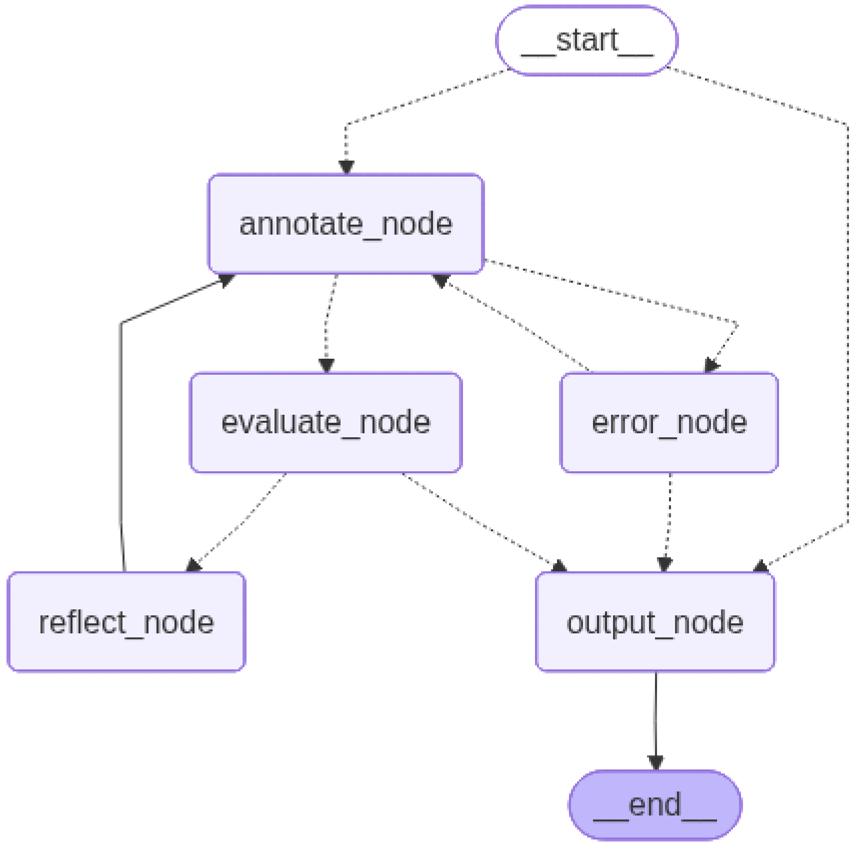

Figure 4.3: *Reflexion LangGraph pipeline architecture with iterative self-critique loop*

The Reflexion graph is entered through a routing function that inspects the task's current confidence level. If the confidence is already HIGH or MEDIUM, the task bypasses the loop entirely and proceeds to the output node. For LOW-confidence tasks, the annotate_node issues an LLM call that includes any prior reflections in the prompt context; on the first trial, no prior reflections exist. The evaluate_node then scores the annotation against the human label using the same F1-based metrics as LLMaAA, and assigns a confidence level that also accounts for the trial number: a perfect score on the first attempt earns HIGH confidence, while the same score on a subsequent attempt earns only MEDIUM, on the grounds that earlier convergence is treated as stronger evidence of reliability.

If the resulting confidence is still LOW and trials remain, the reflect_node generates a natural language self-critique identifying which entities or classifiers were incorrect and why. This reflection is appended to a sliding window of the last three critiques, which is injected into the annotate_node's prompt context on the next trial. The sliding window prevents the prompt from growing unbounded while ensuring the most recent feedback is available. The loop runs for up to three trials before exiting to the output node regardless of whether agreement was reached.

4.5.3 Results and limitations

Reflexion achieved full verification on NLC tasks: all twenty English NLC tasks and all twenty Spanish NLC tasks reached VERIFIED status with mean agreement scores of 0.794 and 0.838 respectively. NER tasks proved substantially more resistant to self-correction. Only thirty percent of English NER tasks were verified, and the average agreement score of 0.358 indicates that even after three trials the model frequently failed to converge on the human annotation. Most tasks exhausted all trials without reaching agreement, with seventy percent of English NER tasks resolved only on the final trial.

The difference in performance between NER and NLC comes down to the nature of the disagreement each task type produces. For NLC, disagreements often arise from genuine uncertainty about which category best applies, a problem that verbal self-critique can address by focusing the model's attention on the distinguishing features of each category. For NER, disagreements frequently reflect structural ambiguity in the label space itself: two or more entity types may be validly applicable to the same span, and no amount of self-critique can resolve a disagreement that originates in schema ambiguity rather than reasoning error. The iterative loop is well-suited to correcting mistakes in reasoning but not to resolving disagreements that are inherent to the annotation schema.

4.6 Active-Synthesis

4.6.1 Motivation and design

Active-Synthesis is this project's contribution to the pipeline design space. It was designed in response to the limitations observed in both Active-Prompt and Reflexion: Active-Prompt identifies uncertain tasks but does not resolve them, while Reflexion attempts resolution through sequential self-critique but struggles when disagreement is structural rather than incidental. Active-Synthesis introduces a second pipeline phase in which all conflicting model outputs are shown simultaneously to the model, which is then asked to produce a single definitive annotation by identifying the conflicts, reasoning from the article text, and justifying its resolution.

The main architectural difference from Reflexion is the synthesis mechanism. Reflexion is a sequential process: the model annotates, evaluates against a known human label, and uses that evaluation to guide its next attempt. Active-Synthesis never shows the model the human label. Instead, the model is given its own conflicting outputs and asked to reason about which elements have consensus, which are genuinely contested, and how to resolve each conflict using evidence from the article. This makes Active-Synthesis applicable even where no clean human label is available, which matters practically given that 98.2% of tasks in this dataset had only a single human annotator.

4.6.2 Graph architecture

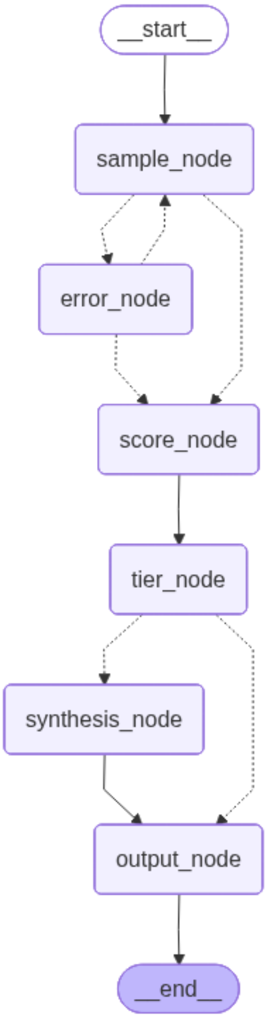

Figure 4.4: *Active-Synthesis Technical Architecture Diagram*

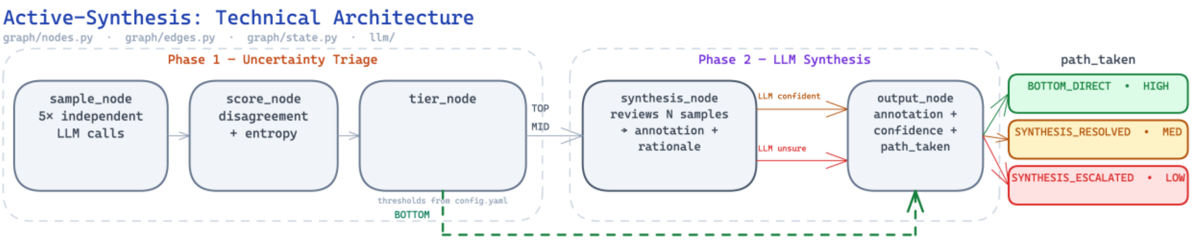

Figure 4.5: *Active-Synthesis complete pipeline graph*

The Active-Synthesis graph reuses the first three nodes of Active-Prompt (sample_node, score_node, and tier_node) with identical logic. The tier assignment step then branches on the triage result. Tasks classified as BOTTOM proceed directly to the output node: when five independent samples agree, no synthesis is needed, and the majority-vote annotation is used as the final output with HIGH confidence. Tasks classified as MID or TOP are routed to the synthesis_node.

The synthesis_node constructs a prompt that presents the article text alongside all five conflicting sample annotations and an explicit count of the conflicting span-type pairs (NER) or classifiers (NLC). The model is instructed to identify where the samples agree, treat those points of consensus as likely correct, and resolve each conflict using textual evidence from the article. The synthesis prompt includes explicit escalation criteria: the model should set a needs_human_review flag if it encounters a conflict it cannot resolve from the text alone, such as a span that was assigned three or more different entity types across samples with no majority. The model's output is a single structured annotation accompanied by a rationale field explaining how each conflict was resolved.

A code-level volume override supplements the model's self-escalation: tasks where the disagreement count exceeds ten for NER or five for NLC are automatically marked for human review regardless of the model's decision, on the grounds that resolving very high-volume conflicts reliably requires domain expertise. This hard threshold prevents the model from overconfidently resolving tasks with a very large number of competing annotations.

4.6.3 Output paths and confidence assignment

The output_node in Active-Synthesis implements four distinct paths based on the task's route through the graph. A task that took the BOTTOM_DIRECT path used the majority vote as its final annotation and is assigned HIGH confidence. A task that was synthesised and resolved without triggering the human review flag follows the SYNTHESIS_RESOLVED path, receiving a final annotation from the synthesis call and MEDIUM confidence. A task where the synthesis call or the code-level override triggered human review follows the SYNTHESIS_ESCALATED path, with the synthesis annotation retained for reference but the confidence set to LOW and the review status set to HUMAN_REVIEW. Tasks where the synthesis call itself failed after retries take the ERROR path, falling back to the majority vote with LOW confidence.

This four-path design ensures that every task exits the pipeline with a structured output regardless of the resolution outcome, which makes the pipeline suitable for batch processing across an entire dataset.

4.6.4 Synthesis prompt design

The synthesis system prompt is substantially longer than the standard annotation prompt, as it must explain both the annotation schema and the synthesis task. It includes the full entity type schema (for NER) or classifier schema (for NLC), the annotation rules from the shared prompt library, and explicit criteria for when to escalate to human review. The user prompt is dynamically constructed to include the article text, the conflict count, and each of the five sample annotations formatted as numbered JSON blocks. This presentation allows the model to compare its own outputs side by side, so the points of agreement and disagreement are legible without additional interpretation.

The synthesis prompt instructs the model to reason from the article text rather than from the samples alone, to keep the synthesis call grounded in the source document. The model is explicitly told that it cannot resolve ambiguity by averaging across samples or deferring to the majority without textual evidence, and that escalation is the appropriate response when textual evidence is insufficient.

4.6.5 Configuration

Active-Synthesis uses the same triage thresholds as Active-Prompt: for English tasks, a disagreement count above three or entropy above 1.5 triggers the synthesis phase; for Spanish tasks, the thresholds are raised to account for greater model variability on that language. The synthesis call uses a larger token budget than the standard annotation calls to accommodate the rationale field and the extended prompt context. All calls are issued at temperature 0.7, consistent with the sampling phase.

4.7 Comparative summary of framework designs

Table 4.1 summarises the four frameworks across the main dimensions of their design.

Table 4.1: Comparative Summary of Framework Designs Across Key Dimensions

Dimension	Active-Prompt	LLMaAA	Reflexion	Active-Synthesis
Primary purpose	Uncertainty triage	Few-shot re-annotation	Iterative self-critique	Conflict resolution
Sampling strategy	N=5 parallel calls; disagreement + entropy computed	Single call per task with KNN demos in context	Single call per trial; prior reflections injected as context	N=5 parallel calls; disagreement + entropy computed (Phase 1)
Resolution mechanism	CoT rationale generated for TOP-tier tasks; no annotation is changed	F1 agreement scored against human label; confidence upgraded if threshold met	Verbal self-critique appended to sliding memory window; re-annotation attempted for up to 3 trials	All N conflicting outputs shown simultaneously to the LLM; produces one definitive annotation with rationale (Phase 2 — unique to this framework)

Uses human label?	No (triage only)	Yes (F1 evaluation)	Yes (agreement scoring)	No (synthesis is blind to human label)
Max LLM calls per task	5 + 1 CoT (TOP)	1 + 5 (uncertainty)	1–3 (loop)	5 + 1 (synthesis)
Applicable to single-annotator tasks?	Yes	Yes	Yes (if human label exists)	Yes
NER performance (EN)	100% TOP — high uncertainty confirmed	0% upgraded (F1 ≈ 0 due to schema)	30% VERIFIED	70.7% SYNTHESIS_RESOLVED
NLC performance (EN)	85% BOTTOM — high LLM agreement	Partial upgrades	100% VERIFIED	100% APPROVED
Primary output	Triage tier + confidence + optional CoT	Upgraded confidence + uncertainty score	Verified confidence + reflexion trace	Final annotation + rationale + path label
Escalation mechanism	TOP tier → HUMAN_REVIEW_HIGH	F1 ≤ 0.5 → remains LOW	Max trials → NEEDS-REVIEW	LLM self-flag + volume override (NER>10, NLC>5)

The frameworks differ most sharply in their relationship to the human label. Reflexion evaluates itself explicitly against the human annotation at each trial, so it is a verification tool rather than an annotation tool. LLMaAA similarly scores its output against the human label. Active-Synthesis deliberately excludes the human label from the synthesis call, so it is a conflict-resolution tool independent of the ground truth, which matters practically given that most tasks in this dataset had only a single annotator.

The design philosophy of each framework determines where it can be applied. Reflexion and LLMaAA work best as validation tools for tasks where a human annotation already exists and can serve as a reference. Active-Synthesis is a resolution tool for tasks where conflicting model outputs must be resolved without reference to an external standard.

Chapter 5: Results and Analysis

5.1 Overview

This chapter reports empirical results for all four annotation frameworks. Active-Prompt, LLMaAA, and Reflexion were each evaluated on twenty tasks per dataset (English NER, English NLC, Spanish NER, Spanish NLC); Active-Synthesis on thirty tasks per dataset for the pilot, then on the full dataset. Human annotations serve as a reference point rather than a definitive quality ceiling — the EDA established that single-annotator coverage and speed-flagged annotations make the human label an incomplete baseline. Sections 5.2 to 5.5 cover each framework in turn; Section 5.6 compares them across NER/NLC performance and treatment of the human label.

5.2 Active-Prompt Results

5.2.1 Triage Distribution

Table 5.1 presents the triage tier distribution, average uncertainty scores, and Chain-of-Thought generation counts for each dataset.

Table 5.1: Active-Prompt Triage Distribution, Uncertainty Scores, and CoT Generation

Dataset	Tasks	TOP (%)	MID (%)	BOTTOM (%)	Avg Disagree	Avg Entropy	CoT Generated
EN NER	20	100.0	0.0	0.0	9.75	4.22	20 (100%)
EN NLC	20	10.0	5.0	85.0	0.85	−0.05*	0 (0%)
ES NER	20	100.0	0.0	0.0	6.60	3.91	20 (100%)
ES NLC	20	0.0	10.0	90.0	1.10	0.08	0 (0%)

The triage results split entirely along task type lines. Every NER task, across both English and Spanish (forty tasks in total), was assigned to the TOP triage tier, meaning every NER task exceeded the disagreement and entropy thresholds for high uncertainty. Chain-of-Thought rationales were generated for all forty NER tasks. NLC tasks showed the opposite pattern: the vast majority fell into the BOTTOM tier, with the five parallel model samples producing near-identical classifications and disagreement and entropy near zero. No CoT rationales were generated for any NLC task.

The average disagreement score for English NER was 9.75 span-type pairs per task, meaning nearly ten entity annotations on average differed between runs across five samples. Spanish NER showed slightly lower disagreement at 6.60, consistent with the shorter average article length in the Spanish dataset observed in exploratory analysis. For NLC, the near-zero disagreement and slightly negative entropy values reflect a distribution so concentrated that the five outputs were essentially identical. The bounded NLC label space puts a natural ceiling on inter-sample variability.

5.2.2 Confidence Mapping

Through the tier-to-confidence mapping, all NER tasks received LOW confidence and all NLC tasks received HIGH or MEDIUM confidence. This means Active-Prompt provides no confident final annotation for any NER task: its output is a triage signal and, for TOP-tier tasks, a CoT rationale, but no definitive annotation to accept or reject. The 100% CoT generation rate for NER reflects the severity of the disagreement; every NER task exceeded the uncertainty threshold, which would be informative for any downstream human review process.

The practical limitation of Active-Prompt is that it diagnoses uncertainty without resolving it. A human reviewer assigned a CoT rationale gains insight into where the model disagreed and why but still receives no definitive annotation. This diagnostic role motivates the three resolution-oriented frameworks that follow.

5.3 LLMaAA Results

5.3.1 Re-Annotation Outcomes

LLMaAA operated on the twenty LOW-confidence tasks per dataset produced by the Active-Prompt triage step. Table 5.2 presents the confidence distribution after re-annotation, the upgrade rate, and the average agreement score with the human label.

Table 5.2: LLMaAA Re-annotation Outcomes: Confidence Upgrade Rate and Agreement Scores

Dataset	Tasks	HIGH (%)	MEDIUM (%)	LOW (%)	Upgrade Rate	Avg Agreement	Outcome
EN NER	20	0.0	0.0	100.0	0%	0.094	All remain LOW
EN NLC	20	5.0	80.0	15.0	85%	0.710	17/20 upgraded
ES NER	20	0.0	0.0	100.0	0%	0.142	All remain LOW
ES NLC	20	5.0	50.0	45.0	55%	0.558	11/20 upgraded

The pattern from Active-Prompt repeats: NER and NLC diverge sharply. For NER, the upgrade rate was zero across both English and Spanish. Every re-annotated NER task remained at LOW confidence, with average agreement scores of 0.094 for English and 0.142 for Spanish, representing near-total misalignment between the LLM's re-annotation and the human label under the F1 metric. For NLC, the pipeline was substantially more effective: 85% of English NLC tasks were upgraded to MEDIUM or HIGH confidence, with an average agreement score of 0.710. Spanish NLC was weaker at 55% upgraded and 0.558 average agreement. The KNN demonstrations retrieved for Spanish articles showed greater variability, which likely contributed to the lower rate.

5.3.2 NER Failure Analysis

The zero upgrade rate for NER is not a failure of the KNN retrieval strategy or the few-shot demonstrations, but a fundamental incompatibility between the evaluation metric and the annotation schema. Span-level F1 requires an exact match on both the span text and the entity type for a pair to count as a true positive. Across a 58-type schema with free-form span boundaries, two independently correct annotations of the same article will typically produce very different sets of span-type pairs, differing on span boundaries, entity type selections where multiple types are valid, and implied entity inclusion decisions. The resulting F1 score between any two valid annotations can approach zero despite both being semantically accurate.

A concrete illustration: for a given article, one annotation might extract a span as "Location Arrest" while another, equally defensible annotation extracts the same span as "Location Reference". Neither is wrong; both reflect legitimate interpretive choices within the schema. Yet F1 counts this as zero true positives for that span, because the type strings do not match exactly. Across a full annotation with ten or more entities, these small divergences compound to produce agreement scores well below the upgrade threshold. Spanish NER also showed maximum uncertainty scores (entropy = 1.0) across all twenty tasks, a consequence of the greater syntactic variability the model must navigate in Spanish-language articles.

A second compounding factor is the comparison target. LLMaAA measures its re-annotation against the original human label, which exploratory analysis confirmed to be systematically incomplete. A re-annotation that correctly extracts ten entities receives a lower F1 score against a two-entity human label than a re-annotation that extracts only the same two entities. The F1-based upgrade gate therefore inadvertently penalises richer, more complete LLM annotations.

5.4 Reflexion Results

5.4.1 Verification Outcomes

Reflexion was applied to the same twenty LOW-confidence tasks per dataset. Table 5.3 presents verification status, average agreement scores, average trials consumed, and the final confidence distribution.

Table 5.3: Reflexion Verification Outcomes: Status, Agreement, and Trial Distribution

Dataset	Tasks	VERIFIED (%)	NEEDS-REVIEW (%)	ERROR (%)	Avg Agreement	Avg Trials	Confidence (H/M/L %)
EN NER	20	30.0	60.0	10.0	0.358	2.55	5 / 25 / 70
EN NLC	20	100.0	0.0	0.0	0.794	1.30	5 / 95 / 0
ES NER	20	55.0	45.0	0.0	0.557	2.50	5 / 50 / 45
ES NLC	20	100.0	0.0	0.0	0.838	1.50	15 / 85 / 0

Reflexion achieved 100% verification on both English and Spanish NLC, with average agreement scores of 0.794 and 0.838 respectively. Most NLC tasks verified on the first trial: 70% for English and 55% for Spanish, giving average trial counts of 1.3 and 1.5. For NLC, the Reflexion loop functions primarily as a confirmation step rather than a correction loop.

For NER, results were substantially weaker. Only 30% of English NER tasks reached VERIFIED status, with the remaining 60% exhausting all three trials and exiting as NEEDS-REVIEW. Spanish NER performed better at 55% verified, though the average agreement score of 0.557 reflects that even verified tasks often fell just above the agreement threshold rather than achieving close alignment with the human label. The trial distribution for English NER shows 70% of tasks requiring all three trials, which is a significant API cost for a pipeline that still cannot verify the majority of tasks.

5.4.2 Self-Correction Behaviour

Going through the reflexion traces, the self-critique mechanism does successfully identify and correct specific annotation errors. One pattern appears across multiple NER tasks: the span "100 years", drawn from prison sentences such as "sentenced to 100 years in prison", being incorrectly assigned the entity type Trafficker Age in an initial annotation. The reflection for such cases consistently identifies this as a contextual misapplication of the numeric entity type, and subsequent trials correctly omit the span. The verbal self-critique mechanism is working as designed; targeted reasoning about specific errors leads to targeted corrections.

This error-correction capability does not, however, translate into high verification rates for NER, because the verification gate is calibrated to the human label rather than to annotation quality in an absolute sense. A task where the model corrects one error but also produces ten additional entities not in the human label will still fail the F1 threshold, even if those additional entities are correct enrichments. The Reflexion loop is effective at reducing false positives within the constraints the model has already identified, but it cannot converge when the disagreement between the LLM's enriched output and the incomplete human label is structural rather than correctable.

5.5 Active-Synthesis Results

5.5.1 Pilot Evaluation (n=30 per dataset)

Table 5.4 presents the path distribution and approval rates for the pilot evaluation.

Table 5.4: Active-Synthesis Pilot Evaluation

Dataset	Tasks	BOTTOM_DIRECT (%)	SYNTHESIS_RESOLVED (%)	SYNTHESIS_ESCALATED (%)	APPROVED (%)	Avg Disagree
EN NER	30	0.0	83.3	16.7	83.3	7.47

EN NLC	30	90.0	10.0	0.0	100.0	0.90
ES NER	30	3.3	83.3	13.3	86.7	7.17
ES NLC	30	93.3	6.7	0.0	100.0	1.07

For NER, 100% of English and 96.7% of Spanish tasks entered Phase 2 via the synthesis node, confirming that structural disagreement is the default condition for NER tasks. The synthesis call resolved 83.3% of English and Spanish NER tasks and produced MEDIUM-confidence annotations for all of them. The remaining 16.7% and 13.3% respectively were escalated to human review; each escalation came with a synthesis rationale explaining the specific conflicts the model could not resolve. No NER task produced an error: all tasks exited with a structured output.

For NLC, the vast majority of tasks never required synthesis. Between 90% and 93.3% of NLC tasks were resolved through the BOTTOM_DIRECT path, with all five samples agreeing sufficiently for the majority vote to serve as the high-confidence final annotation. The small number of NLC tasks that entered Phase 2 were all resolved by the synthesis call, bringing the approval rate for both English and Spanish NLC to 100% with zero human-review escalations.

5.5.2 Full Dataset Evaluation

Table 5.5 presents the full dataset results.

Table 5.5: Active-Synthesis Full Dataset Evaluation: Path Distribution and Approval Rates

Dataset	Tasks	BOTTOM_DIRECT (%)	SYNTHESIS_RESOLVED (%)	SYNTHESIS_ESCALATED (%)	APPROVED (%)
EN NER (full)	215	0.0	70.7	29.3	70.7
EN NLC (full)	217	92.6	7.4	0.0	100.0
ES NER (full)	1,120	0.6	80.5	18.8	81.2
ES NLC (full)	1,120	96.3	3.7	0.0	100.0

The full-dataset results are broadly consistent with the pilot. English NER showed a lower synthesis-resolved rate of 70.7% compared to 83.3% in the pilot, with a correspondingly higher escalation rate of 29.3%. This reflects the greater diversity of article types in the full dataset: more articles with complex multi-party trafficking networks, more schema boundary cases, and more implied entity decisions, all of which increase the probability of triggering the volume or presence conflict escalation criteria. Spanish NER performed comparably to the pilot at 80.5% resolved, with 18.8% escalated.

NLC results at full scale were even stronger: 92.6% of English NLC and 96.3% of Spanish NLC tasks were resolved through the BOTTOM_DIRECT path, confirming that the bounded NLC label space produces near-universal model agreement at volume. Approval rates were 100% for both NLC datasets with zero human-review flags.

5.5.3 Synthesis Rationale Quality

A qualitative review of synthesis rationales for resolved tasks showed consistent use of article evidence to justify annotation decisions. For example, a task where conflicting samples disagreed on whether to label a numeric span as a victim age or as a prison sentence duration was resolved by the synthesis call citing the article's use of "sentenced to" as evidence that the span refers to a legal penalty rather than a biographical detail. Similarly, for tasks where implied entity inclusion was contested across samples, the synthesis rationale identified whether the evidential phrase cited in each sample was genuinely present in the article text, using this as the inclusion criterion.

For escalated tasks, the rationales served a different function: explaining which escalation criteria were triggered and why the conflict could not be resolved from the text. Tasks that escalated on the VOLUME criterion (more than ten conflicting span-type pairs) were accompanied by rationales that still identified the points of consensus and specifically named the contested spans and types requiring expert judgement. Escalated tasks are not returned as blank failures; they come back as structured partial annotations with an attached explanation, which reduces the burden on human reviewers compared to a cold re-annotation request.

5.6 Cross-Framework Comparison

5.6.1 NER Performance

Table 5.6 compares the four frameworks on NER resolution rates for English and Spanish.

Table 5.6: Cross-Framework NER Resolution Rate Comparison

Framework	EN NER Resolved (%)	EN NER Unresolved (%)	ES NER Resolved (%)	ES NER Unresolved (%)	Notes
Active-Prompt	0	100 (all LOW, no annotation)	0	100 (all LOW, no annotation)	Triage only; no resolutio

					n attempted
LLMaAA	0	100 (F1 ≈ 0)	0	100 (F1 ≈ 0)	0% upgrade; F1 metric incompatible with 58-type schema
Reflexion	30	70 (NEEDS-REVIEW)	55	45 (NEEDS-REVIEW)	Self-critique helps; 70% EN exhaust max trials
Active-Synthesis	83	17 (SYNTHESIS_ESCALATED)	87	13 (SYNTHESIS_ESCALATED)	Best NER result; escalations include candidate annotation

The NER results show a clear progression across frameworks. Active-Prompt and LLMaAA both produce zero resolved annotations: the former because it is a triage tool that produces no final annotation, and the latter because the F1-based upgrade criterion is incompatible with the 58-type schema. Reflexion achieves a meaningful improvement through its self-critique loop, resolving 30% of English and 55% of Spanish NER tasks, but is limited by both the human label dependency and the structural nature of NER disagreement. Active-Synthesis achieves the highest resolution rate, resolving more than 80% of NER tasks in both languages by replacing the agreement-against-human-label criterion with an internal conflict resolution mechanism that does not require a ground truth.

5.6.2 NLC Performance

Table 5.7 compares the four frameworks on NLC resolution rates.

Table 5.7: Cross-Framework NLC Resolution Rate Comparison

Framework	EN NLC Resolved (%)	EN NLC Unresolved (%)	ES NLC Resolved (%)	ES NLC Unresolved (%)	Notes
Active-Prompt	85	15 (MID/TOP)	90	10 (MID)	BOTTOM tasks = HIGH confidence; no synthesis needed
LLMaAA	85	15 (remain LOW)	55	45 (remain LOW)	Good EN result; ES NLC weaker due to lower agreement
Reflexion	100	0	100	0	100% VERIFIED; most verify on Trial 1
Active-Synthesis	100	0	100	0	90–96% BOTTOM_DIRECT; zero human-review flags

For NLC, three of the four frameworks achieve strong results. Active-Prompt correctly identifies 85–90% of NLC tasks as high-confidence based on low inter-sample disagreement. Reflexion and Active-Synthesis both achieve 100% resolution. LLMaAA is the weakest NLC performer, leaving 15–45% of tasks at LOW confidence; the F1 threshold does not account for the minor category-level disagreements that even well-performing NLC annotations exhibit. The near-perfect NLC performance of Reflexion and Active-Synthesis reflects the fundamental tractability of the task: when the label space is bounded and model outputs are consistent, any resolution mechanism converges reliably.

5.6.3 The NER/NLC Asymmetry

The consistent gap between NER and NLC performance across all four frameworks points to a structural property of the annotation problem rather than a failure of any specific pipeline. NLC has a label space of approximately forty possible classifier-category combinations, with well-defined semantic boundaries between categories. NER has a combinatorial label space that grows with article length and entity density: for an article with fifteen relevant entities, each assignable to one of 58 types with variable span boundaries, the number of possible valid annotations is extremely large. The model's five parallel samples explore different regions of this space on each run, which generates high disagreement as a natural consequence of the schema's size, not as a sign of low model quality.

This matters for how results should be evaluated. F1-based agreement, which requires exact string and type matches, is ill-suited to NER annotation evaluation when the schema is large and span boundaries are flexible. The only framework that avoids this problem entirely is Active-Synthesis, which does not use F1 at any point and instead uses the model's own reasoning about article evidence as the quality criterion. The strong NER results for Active-Synthesis are partly a consequence of using an evaluation criterion that fits the task type

Chapter 6: Discussion

6.1 Overview

This chapter interprets the Chapter 5 results against the broader research questions, the literature from Chapter 2, and the practical constraints of the project. Four main threads run through it. First, the consistent NER/NLC asymmetry observed across all frameworks is treated as a structural property of the annotation problem rather than a pipeline-specific artefact, with implications for how annotation quality should be measured in high-cardinality entity schemas. Second, the limitations of using human annotations as the quality gate for LLM outputs are examined against the realities of this specific dataset, where single-annotator coverage, speed-flagged annotations, and systematic incompleteness create a baseline that is itself unreliable. Third, Active-Synthesis is situated as a novel contribution to the annotation enhancement literature, with its design choices examined in relation to the three frameworks it builds upon. Finally, the study's limitations are discussed, covering sample sizes, model choice, and the absence of an independent expert evaluation.

6.2 The NER/NLC Asymmetry as a Structural Property

The most consistent finding across all four frameworks is the complete divergence in behaviour between Named Entity Recognition and Natural Language Classification tasks. Every framework that worked well for NLC (Active-Prompt with its high BOTTOM rate, Reflexion at 100% verified, Active-Synthesis at 100% approved) showed substantially weaker results for NER, while LLMaAA failed entirely on NER. That sequence labeling tasks are harder than bounded text classification is well established in NLP; the contribution of this section is to characterise the specific mechanism by which that asymmetry resists resolution across four structurally different annotation frameworks in a high-cardinality specialist schema, and to identify what this implies for annotation quality measurement and framework design. This consistency across four architecturally different frameworks strongly suggests the asymmetry is not a pipeline design issue but a fundamental property of the two task types.

The source of the asymmetry is the structure of the label space. The NLC schema presents the model with six classifiers, each drawing from a bounded set of approximately five to eight categories. The model's task on any given article is to select from a relatively small set of predefined options. With well-separated semantic categories and clear article-level relevance signals, the model converges on the same classification across independent runs, producing the near-zero disagreement scores observed consistently. The NER schema, by contrast, presents the model with 58 entity types and requires two simultaneous decisions per entity: what text to extract as a span, and which of 58 types best describes it. With free-form span boundaries and multiple plausible type assignments for many spans, the combinatorial space of valid annotations is orders of magnitude larger than for NLC. Independent runs explore different regions of this space; the structural disagreement they generate is a property of the annotation task itself, not a failure of the model.

This matters for how annotation quality should be defined and measured in NER contexts. The frameworks that attempt to measure quality by comparing LLM output against an existing annotation, apply a criterion that presupposes a unique correct annotation. For NLC, this presupposition is largely valid because the bounded label space makes convergence natural. For NER, it is not: two independently valid annotations of the same article may share very few span-type pairs in common, not because either is wrong, but because the label space admits many valid interpretations. The LLMaAA and Reflexion results show that evaluation metrics designed for bounded classification tasks are inappropriate for high-cardinality entity recognition. Any future annotation quality framework for NER should account for the multiplicity of valid annotations, whether through a relaxed matching criterion, semantic similarity rather than exact string matching, or a label-free quality criterion such as the internal conflict resolution mechanism used by Active-Synthesis.

6.3 Rethinking the Human Label as Ground Truth

Running through the entire results chapter is a related issue: the role of the human annotation as a reference standard. LLMaAA and Reflexion both treat the human label as a quality ceiling; an LLM annotation is only upgraded if it agrees with the human label beyond a threshold. This design is well-motivated in the general annotation quality literature, where the goal is to produce annotations that faithfully represent human judgement. In this project, however, the exploratory data analysis in Chapter 3 established that the human label cannot serve this role reliably.

The EDA revealed that 98.2% of tasks had only a single annotator (no consensus signal), 969 labels were speed-flagged for sub-two-second completion, and per-annotator reliability varied substantially across the pool. To conclude, the human label is a noisy, incomplete reference rather than a reliable gold standard.

The consequences for LLM-based annotation enhancement are real. When a human captures two entity spans and the LLM correctly identifies ten, F1 against the human label will be low not because the LLM is wrong, but because the reference is incomplete. This creates a systematic bias: frameworks that gate quality on human agreement penalise enrichment, which is exactly what LLM-based annotation enhancement is meant to provide. The LLMaAA 0% NER upgrade rate and Reflexion's NEEDS-REVIEW accumulation both reflect this directly.

Active-Synthesis avoids this problem by design: it never consults the human label during the resolution process. The human annotation is stored in the task state and available for downstream use, but it plays no role in determining whether the synthesis output is accepted or escalated. The quality criterion is internal: can the model resolve its own disagreements using evidence from the article text? This is both appropriate for a single-annotator dataset and aligned with the enrichment objective. This is a design choice, not a workaround: in the absence of a reliable ground truth, internal consistency across model outputs is a more valid quality signal than agreement with a single noisy annotation.

6.4 Situating Active-Synthesis in the Literature

6.4.1 Relationship to Active-Prompt

Active-Synthesis shares Phase 1 entirely with Active-Prompt: the same parallel sampling, disagreement and entropy scoring, and tier assignment logic. Active-Prompt's results confirm the uncertainty signal is accurate. Every NER task registers as high-uncertainty, every NLC task as low. What Active-Prompt lacks is any mechanism to act on that signal: its CoT rationale documents the conflict but produces no final annotation. Active-Synthesis completes the architecture by routing MID and TOP tasks to a synthesis phase, transforming the identified conflict into the input material for Phase 2.

6.4.2 Relationship to Reflexion

Both Reflexion and Active-Synthesis attempt to improve an initial annotation through a secondary reasoning step, but differ structurally. Reflexion generates a verbal self-critique against the human label and re-annotates sequentially. Active-Synthesis presents all conflicting samples simultaneously and adjudicates in one call.

6.4.3 Relationship to LLMaAA

LLMaAA and Active-Synthesis are both trying to get reliable annotations out of a model you can't fully trust. LLMaAA feeds the model semantically retrieved examples, betting that similar articles will reduce output variability. Active-Synthesis uses internal context instead: the model's own conflicting outputs from Phase 1 become the thing it reasons over in Phase 2.

6.6 Limitations

6.6.1 Sample Size

The pilot evaluation for Active-Prompt, LLMaAA, and Reflexion was conducted on twenty tasks per dataset, and for Active-Synthesis on thirty tasks per dataset. While Active-Synthesis was also evaluated on the full pre-processed dataset, the comparative evaluation across all four frameworks used small samples. The results are directionally consistent across both languages and task types, but the specific percentages should be treated as indicative rather than definitive; the Reflexion figure of 30% VERIFIED for English NER, for example, rests on just six tasks. A larger comparative evaluation on the full dataset would provide more statistically robust estimates of each framework's performance.

6.6.2 Single Model

All four frameworks were evaluated using GPT-4o at a temperature of 0.7. The choice of GPT-4o was motivated by its strong multilingual capabilities and robust instruction following, which are both important for a 58-type entity schema with complex annotation rules. However, the results are not necessarily generalisable to other large language models. Different models may exhibit different levels of NER disagreement across parallel samples, different self-critique effectiveness in the Reflexion loop, and different synthesis quality. The finding that NER disagreement is structurally high is likely to hold across models given the combinatorial nature of the label space, but the specific resolution rates achieved by Active-Synthesis may vary with model capability and instruction following quality.

6.6.3 Absence of Expert Evaluation

This study evaluates annotation quality primarily through framework-level metrics: triage tiers, confidence levels, verification rates, and agreement scores. For Active-Synthesis, the synthesis rationale provides a qualitative basis for assessing resolution quality, and the SYNTHESIS_RESOLVED / SYNTHESIS_ESCALATED distinction captures the model's own confidence in its outputs. However, this study did not conduct an independent expert evaluation of the produced annotations. A gold standard evaluation in which domain experts reviewed a sample of SYNTHESIS_RESOLVED annotations against the article text would give stronger evidence of quality than the internal consistency metrics used here. This is a significant gap, and expert review of a sample of synthesis outputs should be the next step.

6.6.4 Threshold Calibration

The triage thresholds used by Active-Prompt and Active-Synthesis (the disagreement and entropy values that determine tier assignment) were set empirically through inspection of the pilot data rather than through systematic calibration against a held-out evaluation set. Similarly, the escalation criteria in the synthesis prompt (VOLUME > 10 for NER, PRESENCE < 3/5, etc.) were designed based on inspection of the annotation disagreement patterns rather than derived from optimisation. Calibrating these thresholds against expert-validated annotations could improve the precision of the triage and escalation decisions, potentially increasing the SYNTHESIS_RESOLVED rate or reducing false escalations.

6.6.5 Language Scope

The evaluation covers English and Spanish, the two primary languages in the Stop the Traffik dataset. The project's use of a multilingual sentence transformer for the LLMaAA demo pool and language-specific triage thresholds for Active-Prompt and Active-Synthesis reflects a commitment to multilingual operation. However, the frameworks were not tested on other languages present in the dataset, or on MSHT datasets in less well-resourced languages. GPT-4o's multilingual capabilities are strong but uneven across languages, and the elevated disagreement scores observed for Spanish NER relative to English suggest that model performance degrades somewhat on the Spanish data.

6.7 Ethical Considerations

The articles processed by these pipelines describe real victims, real perpetrators, and ongoing cases. The article text was passed to third-party API calls unchanged, meaning sensitive case details were not filtered before LLM processing. In a domain where victim safety and live investigations are on the line, that's worth acknowledging as a limitation of the current implementation.

There's also a bias risk that this project hasn't fully resolved. Models trained on internet-scale text carry assumptions about trafficking patterns, victim demographics, and perpetrator profiles that may not reflect the global MSHT context at all. The synthesis rationale and escalation mechanism add some transparency, but a demographic audit of the entity extractions should happen before any of this feeds into large-scale analysis.

The outputs may eventually shape resource allocation or policy recommendations. The confidence tier system is explicit about reliability, but SYNTHESIS_ESCALATED annotations shouldn't go into analytical models without a domain expert reviewing them first.

Chapter 7: Conclusion

7.1 Summary

This project, carried out in partnership with Stop the Traffik, looked at whether LLM-based frameworks could improve annotation quality on a dataset of modern slavery and human trafficking news articles. The dataset reflected real-world conditions honestly: 98.2% of tasks had a single annotator, mean inter-annotator agreement sat at 26.1%, and exploratory analysis turned up speed-flagged annotations and unreliable annotators. Standard agreement metrics didn't apply, so the project needed a different quality framework.

Four pipelines were built and evaluated across English and Spanish NER and NLC tasks: Active-Prompt, LLMaAA, Reflexion, and Active-Synthesis. Active-Synthesis was the novel contribution. It combined uncertainty-based parallel sampling with a conflict-resolution phase where all conflicting outputs were handed back to the model at once, which then produced a single annotation justified by evidence from the article. All four pipelines ran on LangGraph with shared infrastructure, using GPT-4o at temperature 0.7.

7.2 Key Findings

The NER/NLC asymmetry was the most consistent finding across all frameworks. NLC tasks, which use a bounded six-classifier schema, produced near-zero inter-sample disagreement and were tractable for every approach. NER tasks, with 58 entity types and free-form span boundaries, produced structurally high disagreement that is a property of the label space rather than a model failure. This asymmetry persisted regardless of which framework was applied, confirming it is a structural feature of the annotation problem rather than a pipeline artefact.

Agreement-based evaluation metrics proved inappropriate for the NER schema. LLMaAA's F1-based upgrade criterion and Reflexion's agreement-based verification both presuppose a unique correct annotation, which does not hold across a 58-type schema where valid annotations of the same article may share few span-type pairs. Both frameworks also treated the incomplete human labels as a reliable quality ceiling, penalising LLM enrichment rather than rewarding it. Active-Synthesis avoided both problems by using an internal conflict-resolution criterion that does not reference the human label at any stage.

Active-Synthesis achieved the strongest results overall. It approved 83–87% of NER tasks in the pilot evaluation and 70–81% at full dataset scale, with 100% NLC approval and zero human-review flags across all NLC datasets. Its SYNTHESIS_ESCALATED outputs, produced for the unresolved minority, include a candidate annotation and a rationale identifying the specific conflicts. Human reviewers receive a structured brief rather than a blank task. Table 7.1 ranks all four frameworks.

Table 7.1: Overall Framework Performance Rankings

Rank	Framework	NER Result	NLC Result	Key Strength	Key Limitation
1	Active-Synthesis	83–87% approved	100% approved	Label-free conflict resolution; structured output for every task	Higher API cost; no expert evaluation yet
2	Reflexion	30–55% verified	100% verified	Self-critique corrects specific reasoning errors	Fails on structural NER disagreement; human label dependency
3	Active-Prompt	Triage only (100% TOP)	Triage only (85–90% BOTTOM)	Accurate uncertainty signal; CoT rationale for reviewers	Produces no final annotation
4	LLMaAA	0% upgraded	55–85% upgraded	KNN retrieval sound; strong English NLC result	F1 incompatible with 58-type NER schema; human label dependency

7.3 Novel Contribution

Active-Synthesis is the primary novel contribution of this dissertation. No prior work known to the author combines uncertainty-based parallel sampling with a simultaneous multi-sample synthesis call for annotation conflict resolution. The framework makes two conceptual contributions to the annotation quality literature. First, it demonstrates that treating LLM inter-sample disagreement as the input to a resolution process (rather than as a signal of failure) is more effective for structurally ambiguous tasks. Second, it demonstrates that a label-free quality criterion, grounded in the model's ability to resolve its own conflicts using article evidence, is both valid and practical for datasets where the human baseline is incomplete or unreliable. A secondary contribution is the three-tier confidence classification system (HIGH/MEDIUM/LOW) derived from a weighted combination of inter-annotator agreement, annotator count, and lead-time penalty, which served as the consistent quality framework across the entire project.

7.4 Recommendations and Future Work

Three recommendations follow for Stop the Traffik. First, deploy Active-Synthesis as the primary annotation enhancement pipeline. Process NLC tasks first, where nearly all will resolve through the BOTTOM_DIRECT path at minimal API cost, before moving to NER. Second, commission an independent expert review of a sample of SYNTHESIS_RESOLVED NER annotations to validate pipeline quality before using the outputs as training data for downstream models. Third, review the NER schema design using the escalation rationales as a diagnostic: the types most frequently contested in synthesis outputs are candidates for clarification or consolidation.

The most pressing future directions are expert evaluation, compact model integration, and metric development. Once a validated pool of MEDIUM-confidence Active-Synthesis annotations exists, it could serve as training data for a GLiNER-style compact model fine-tuned on the 58-type MSHT schema, which would enable lower-cost inference at scale. In parallel, developing a relaxed NER evaluation metric that awards partial credit for type-approximate matches would provide a more meaningful quality signal than exact-match F1 for future framework comparisons. A hybrid approach combining Active-Synthesis's conflict resolution with Reflexion's self-critique for escalated tasks is also worth exploring, as the synthesis rationale provides richer starting context than a standard reflection prompt.

7.5 Closing Remarks

This study demonstrated that LLM-based annotation enhancement is feasible and effective for a specialist humanitarian NLP task, provided the framework design fits the annotation schema's structure and accounts honestly for the reliability of the available human labels. The failure of agreement-based metrics on the 58-type NER schema, and the success of label-free synthesis, point to a clear lesson: quality frameworks must be designed with the label space in mind, not adapted from adjacent tasks where the structure differs. Active-Synthesis can produce richer, more consistent MSHT annotations without requiring prior labelled data, model training, or crowdsourcing infrastructure. For humanitarian organisations where specialised schema expertise is scarce and annotation budgets are limited, that is a real practical difference.

References

Alhazmi, K. and Alsumari, W. (2021) 'Effects of annotation quality on model performance', *2021 International Conference on Artificial Intelligence in Information and Communication (ICAIIC)*. IEEE. doi: 10.1109/ICAIIC51459.2021.9415271.

Ash, J.T., Zhang, C., Krishnamurthy, A., Langford, J. and Agarwal, A. (2020) 'Deep batch active learning by diverse, uncertain gradient lower bounds', *International Conference on Learning Representations (ICLR 2020)*.

Bansal, P. and Sharma, A. (2023) 'Large language models as annotators: Enhancing generalization of NLP models at minimal cost', *arXiv preprint arXiv:2306.15766*.

Bernhardt, M., Castro, D.C., Tanno, R., Schwaighofer, A., Tezcan, K.C. and Monteiro, M. (2022) 'Active label cleaning for improved dataset quality under resource constraints', *Nature Communications*, 13(1). doi: 10.1038/s41467-022-28818-3.

Yazdani, A., Stepanov, I. and Teodoro, D. (2025) 'GLiNER-BioMed: A suite of efficient models for open biomedical named entity recognition', *arXiv preprint arXiv:2504.00676*.

Diao, S., Wang, P., Lin, Y., Pan, R., Liu, X. and Zhang, T. (2023) 'Active prompting with chain-of-thought for large language models', *arXiv preprint arXiv:2302.12246*.

Dong, Y., Jiang, L. and Li, C. (2021) 'Improving data and model quality in crowdsourcing using co-training-based noise correction', *Information Sciences*, 583, pp. 174–189. doi: 10.1016/j.ins.2021.11.015.

Du, Y., Li, S., Torralba, A., Tenenbaum, J.B. and Mordatch, I. (2023) 'Improving factuality and reasoning in language models through multiagent debate', *arXiv preprint arXiv:2305.14325*.

Gilardi, F., Alizadeh, M. and Kubli, M. (2023) 'ChatGPT outperforms crowd workers for text-annotation tasks', *Proceedings of the National Academy of Sciences*, 120(30). doi: 10.1073/pnas.2305016120.

Li, M., Fahlstrom Myrman, A., Mu, T. and Ananiadou, S. (2019) 'Modelling instance-level annotator reliability for natural language labelling tasks', *arXiv preprint arXiv:1905.04981*.

Lu, Y. and Smith, K. (2025) 'Feeding LLM annotations to BERT classifiers at your own risk', *arXiv preprint arXiv:2504.15432*.

Madaan, A. et al. (2023) 'Self-Refine: Iterative refinement with self-feedback', *arXiv preprint arXiv:2303.17651*.

Meguellati, E., Zeghina, A., Sadiq, S. and Demartini, G. (2025) 'LLM-based semantic augmentation for harmful content detection', *arXiv preprint arXiv:2504.15548*.

Rodrigues, F., Pereira, F. and Ribeiro, B. (2013) 'Learning from multiple annotators: Distinguishing good from random labelers', *Pattern Recognition Letters*, 34(12), pp. 1428–1436. doi: 10.1016/j.patrec.2013.05.011.

Sener, O. and Savarese, S. (2018) 'Active learning for convolutional neural networks: A core-set approach', *International Conference on Learning Representations (ICLR 2018)*.

Shinn, N., Cassano, F., Labash, B., Gopinath, A., Narasimhan, K. and Yao, S. (2023) 'Reflexion: Language agents with verbal reinforcement learning', *Advances in Neural Information Processing Systems (NeurIPS 2023)*.

Su, B., Jiang, L. and Si, S. (2025) 'Confident learning-based noise correction for crowdsourcing', *Pattern Recognition*. doi: 10.1016/j.patcog.2025.111622.

Sylolypavan, A., Sleeman, D., Wu, H. and Sim, M. (2023) 'The impact of inconsistent human annotations on AI driven clinical decision making', *npj Digital Medicine*, 6(1). doi: 10.1038/s41746-023-00773-3.

Tan, Z. et al. (2024) 'Large language models for data annotation and synthesis: A survey', *arXiv preprint arXiv:2402.13446*.

Qin, M., Zhu, R., Xia, M., Chen, C., Zhu, Z., Lin, M., Zhao, J., Xu, L., Fan, C., Wu, R. and Wang, H. (2025) 'CrowdAgent: Multi-agent managed multi-source annotation system', *arXiv preprint arXiv:2509.14030*.

Wang, X., Chi, X., Song, Y. and Yang, Z. (2023) 'Active learning with label quality control', *PeerJ Computer Science*, 9, e1480. doi: 10.7717/peerj-cs.1480.

Wang, Z., Boix, X., Bhaskaran, K., Hamid, R. and Gao, Y. (2021) 'A review and experimental analysis of active learning over crowdsourced data', *Artificial Intelligence Review*, 54(7), pp. 5283–5305. doi: 10.1007/s10462-021-10021-3.

Xiao, R. et al. (2023) 'FreeAL: Towards human-free active learning in the era of large language models', *Proceedings of EMNLP 2023*, pp. 14520–14535.

Yao, S. et al. (2024) 'Graph of thoughts: Solving elaborate problems with large language models', *Proceedings of AAAI 2024*.

Zhang, R., Li, Y., Ma, Y., Zhou, M. and Zou, L. (2023) 'LLMaAA: Making large language models as active annotators', *Findings of ACL: EMNLP 2023*.